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Enhancing the Business Sustainability of Automotive Glass

Assessing Environmental, Social and Value Chain Impacts of Automotive Glass 2026

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Abstract

This thesis focuses on the automotive glass industry and examines the comparison between sustainability-oriented innovation management theory and its practical application, with particular emphasis on sustainability-related business risks within the value chain. As vehicles become increasingly electrified, emissions from the use phase are expected to decrease significantly, particularly when renewable energy is used for charging. In contrast, emissions associated with raw materials have increased, due to the improvement of batteries capacities, and are becoming a major contributor to the overall life-cycle emissions of vehicles. Although glass does not represent the largest material share in a vehicle, improvements in the sustainability performance of other major materials have elevated automotive glass as a more significant area of concern. The methodology applied in this research includes a literature review, semi-structured stakeholder interviews, an environmental, social, governance and business risk assessment based on the ESSENZ+ framework, a sustainability failure mode and effects analysis, and a materiality assessment to prioritize mitigation actions. Through this study, the automotive glass value chain was systematically mapped, and key sustainability-related risks were identified. In addition, Sustainable Oriented Innovation practices were analyzed, leading to the development of a structured framework aimed at guiding Original Equipment Manufacturers and suppliers toward more sustainable automotive glass systems. This framework supports improved traceability of raw materials, enhanced circularity, and increased collaboration across the automotive glass industry.

Keywords

Automotive Glass, Value Chain, Sustainable Oriented Innovation, Sustainable Sourcing, ESG Risk Assessment, Circular Economy, ESSENZ+ method

Sammanfattning

Denna rapport fokuserar på fordonsglasindustrin och undersöker jämförelsen mellan teorin om hållbarhetsorienterad innovation och dess praktiska tillämpning, med särskilt fokus på hållbarhetsrelaterade affärsrisker inom värdekedjan. I takt med att fordon i allt större utsträckning elektrifieras förväntas utsläppen från användningsfasen minska avsevärt, särskilt när förnybar energi används för laddning. Däremot har utsläppen kopplade till råmaterial ökat och utgör en allt större andel av fordonens totala livscykelutsläpp. Även om glas inte representerar den största materialandelen i ett fordon, har förbättringar i hållbarhetsprestandan hos andra huvudsakliga material medfört att fordonsglas har blivit ett mer betydande område ur hållbarhetssynpunkt. Den metodik som tillämpas i denna studie omfattar en litteraturöversikt, semistrukturerade intervjuer med intressenter, en riskbedömning med avseende på miljö, socialt ansvar och bolagsstyrning riskbedömning baserad på ESSENZ+-ramverket, en analys av en FMEA och effekter med fokus på hållbarhet samt en materialitetsbedömning för att prioritera åtgärder. Genom studien har värdekedjan för fordonsglas systematiskt kartlagts och centrala hållbarhetsrelaterade risker identifierats. Vidare har praktiker hållbarhetsorienterad innovation analyserats, vilket resulterat i utvecklingen av ett strukturerat ramverk avsett att vägleda originaltillverkare och leverantörer mot mer hållbara system för fordonsglas. Detta ramverk främjar förbättrad spårbarhet av råmaterial, ökad cirkularitet samt stärkt samarbete inom fordonsglasindustrin.

Nyckelord

Fordonsglas, Värdekedjan för fordonsglas, Hållbarhetsinriktad innovation, Hållbar anskaffning av resurser, ESG-riskbedömningar, Cirkulär ekonomi, ESSENZ+ metod

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From Mohamid Alfarraj: *I would like to take this chance to thank my family (Asmaa, Hassan, Sara and Sidra) whom have supported me during my studies and friends which have inspired me along the journey.*

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Furthermore, throughout the thesis, AI tools were used to support grammatical accuracy and improve the clarity and alignment of the text. These tools were not used to generate content, but rather to refine and correct text written by the authors.

Stockholm, June 2026

Mohamid Alfarraj Xiomara N. Caipo Dominguez

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Listings

List of Acronyms

AGRSS	Automotive Glass Replacement Safety Standard
AGSC	Auto Glass Safety Council
ANSI	American National Standards Institute
CCC	China Compulsory Certification
ECE	Economic Commission for Europe
ELV	End-of-Life Vehicle
ESSENZ+	Enhanced Social and Environmental Resource Efficiency Assessment
ESG	Environmental, Social, and Governance
FMEA	Failure Mode and Effects Analysis
GSCM	Green Supply Chain Management
GRI	Global Reporting Initiative
ILO	International Labour Organization
IRMA	Initiative for Responsible Mining Assurance
KPI	Key Performance Indicator
LCA	Life Cycle Assessment
NGO	Non-Governmental Organization
OEM	Original Equipment Manufacturer
PFAS	Per- and Polyfluoroalkyl Substances
PVB	Polyvinyl Butyral
RMoC	Raw Material of Concern
RPN	Risk Priority Number
S-FMEA	Sustainable Failure Mode and Effects Analysis
SOI	Sustainability-Oriented Innovation
UN	United Nations
USGS	United States Geological Survey

Chapter 1

Introduction

This chapter introduces the project background, purpose and goals; as well as delimitation of the thesis project.

1.1 Background

In recent years, the focus on sustainability and the circular economy has steadily increased, particularly following the 2015 Paris Agreement. The agreement aims to reduce global CO_2 emissions in order to limit global warming to 1.5°C above pre-industrial levels and to decrease 43% of greenhouse gas emissions by 2030 [1]. Since its implementation, national legislation has increased and has become one of the primary drivers of innovation for many global organizations [2].

Other important drivers of sustainability have been identified in its significance for business. Customers are becoming increasingly aware of the impacts of products and are therefore demanding ethical and sustainable practices in their development. In addition, there are direct benefits for businesses. Developing in a sustainable manner is considered advantageous for the long-term viability of a company. Purchasing recycled materials can be more cost-efficient and environmentally friendlier compared to the use of primary materials in most cases.

Finally, adopting a circular approach addresses the global trend in which regionalization is increasingly replacing globalization, making access to raw materials more critical for businesses [3]. Consequently, companies have placed greater emphasis on transforming their operations from the traditional

linear model of consumption “take, make, waste” to more circular approaches that minimize environmental impact and strengthen their business position [2].

For the specific case of the automotive industry, according to Statista in 2022, this industry sector was responsible of approximately 21% of total global CO_2 emissions, within that sector, passenger vehicles and vans were responsible for approximately 48% of the total sum of CO_2 emissions [4]. These emissions arise from multiple stages of the vehicle life cycle, including material extraction, production and refining, vehicle manufacturing, use phase, and end-of-life treatment [4]. As the automotive industry transitions from internal combustion engines to electrified powertrains, emissions associated with materials have increased due to larger battery sizes, which affect both vehicle weight and the use of materials with higher CO_2 intensity [4]. At the same time, emissions during the use phase have been significantly reduced, particularly when vehicles are charged using renewable energy, as illustrated in Figure 1.1 [4].

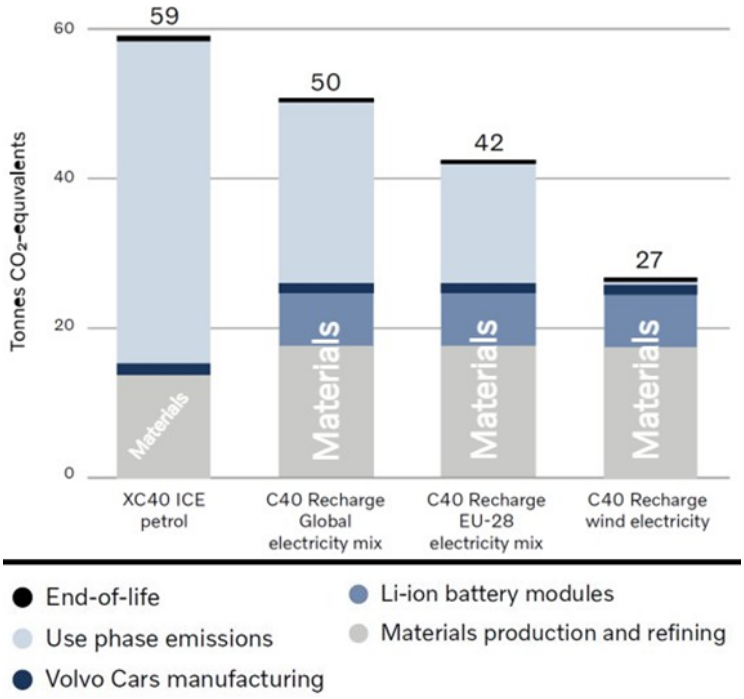


Figure 1.1: *Volvo cars comparison of total tonnes Co2 equivalents between XC40 and C40 models provided by Volvo Cars [4].*

Materials are the core of every product as they critically affect carbon

footprints, price and have significant social impacts on societies. Therefore, issues such as material scarcity and ethical extractions among others are important for long term thrive and resilience of the business [5]. Responsible sourcing and refining of core materials therefore remains of high importance to achieve circularity and true sustainability considering environmental, social and business perspectives [5].

1.1.1 Organization context

Volvo Cars Group is a global company founded in 1927 and owned by Geely corporation since 2010, Today, it is one of the most well-known and respected car brands in the world with sales to customers in more than 100 countries. In 2025, Volvo Cars sold over 710 thousand cars, with an electrified share of 46%. Volvo Cars' head office, product development, marketing and administration functions are mainly located in Gothenburg, Sweden. Additionally, its production plants are located in Gothenburg, Ghent (Belgium), South Carolina (US), Chengdu, Daqing and Taizhou (China). The company also has R&D and design centers in Gothenburg and Shanghai (China) [6].

Volvo Cars designs and manufactures luxury passenger vehicles with a strong focus on passenger safety and sustainability as they acknowledges the environmental impacts to which its vehicles contribute [7]. "For life. To provide freedom to move in a personal, sustainable and safe way." This purpose is reflected in Volvo Cars' ambition to become a fully electric car maker and in its commitment to an ongoing reduction of its carbon footprint, with the ambition to achieve net-zero greenhouse gas emissions by 2040, becoming a fully circular business by 2040 and achieve cost savings of 1 BSEK by 2025 [6, 8, 9]. These targets are considered achievable through design of vehicles with a focus on value retention, resource efficiency, and waste elimination, achieved by optimizing the use of products and components [8, 10].

1.1.2 Sustainability and Innovation Challenges in the Automotive Glass Industry

According to Statista, passenger vehicle production has steadily increased in recent years, with an estimated 93 million vehicles produced in 2024 [11].

On the other hand the proportion of vehicles reaching their end of life is estimated to be approximately 4% annually. This corresponds to around 6 million vehicles becoming waste each year, representing approximately 186 million tons of materials, assuming an average vehicle weight of 2 tons [12, 13]. Given this growth in production and the substantial volume of material waste generated annually, material value recovery from end-of-life vehicles is considered to be of significant importance to reduce material environmental impact [13].

However, glass material recovery and circularity, particularly in the case of automotive glass, remain almost non-existent, as only approximately 1-8% of post-consumer glass is recycled back into production of newer glass components, while the remainder is either downgraded, discarded as waste or sent to landfills [14]. Another important factor to consider is the specific structure of laminated glass, which is commonly used in vehicle windshields and constitutes a significant share of end-of-life vehicle waste. This type of glass presents considerable recycling challenges due to the presence of polyvinyl butyral (PVB) interlayers used to bond the float glass layers [13]. The use of laminated glass in automotive windshields is mandated by UN/ECE Regulation No. 43 due to its superior safety performance characteristics. Laminated glass, bonded with PVB interlayers, provides enhanced safety for the driver and front-seat passengers by maintaining structural integrity and keeping the glass layers bonded together. This prevents the glass from shattering into sharp fragments upon impact and improves occupant protection by reducing the risk of object penetration into the vehicle during collisions [15].

From an Original Equipment Manufacturer (OEM) perspective, glass represents between from 40 to 80 kg of their weight per car depending on the model [16]. At the same time, it is important to highlight that automotive glass industry components are designed and developed under safety requirements during their use, implying different considerations for the material used and its material performance [17]. Additionally the automotive glass manufacturing process also present significant environmental impacts due to its extensive use of energy, high extraction of raw materials such as soda ash and sand silica and low rate of recovery [18, 19]. Automotive glass therefore represents a great opportunity to improve the sustainability performance of passenger vehicles.

Innovation practices within the automotive glass sector are of particular

interest, as the industry has not experienced any major technological production breakthroughs since the introduction of the float glass process by Pilkington in 1952 [20]. Instead, the sector has been characterized by an increasing number of incremental innovations, both in technical aspects such as the development of smart glass integrating electronic components and in sustainability-related improvements, including the electrification of furnaces and recycling initiatives in float glass production [21, 22].

While sustainability and innovation are often considered complementary drivers of long-term competitive advantage for organizations, in practice, the most commonly implemented sustainability actions focus on optimizing the exploitation of existing resources [23, 24]. However, theory suggests that greater emphasis should be placed on more radical approaches that prioritize the exploration of new solutions. Rather than solely improving recycling methods, this perspective advocates for the development of entirely new and sustainable business models [24].

In this context, the automotive glass industry represents a particularly relevant area in which to investigate innovation practices. As noted by Vollenbroek (2002), truly sustainable innovations require significant environmental, social, and economic impacts, as well as organizational change and increased cross-sector collaboration to enable a transition toward more sustainability-oriented solutions [23]. Similarly, Arekrans et al. (2023) highlight that a lack of cross-functional and cross-divisional coordination tends to reinforce exploitative organizational behaviors rather than exploratory ones. They argue that such continuous practices risk leading organizations to improve existing processes without addressing fundamental challenges, thereby "doing the wrong things better." Consequently, they emphasize the importance of balancing exploration and exploitation in order to achieve meaningful and radical long-term change [24].

1.2 Research Gap, Purpose & Goals

While Life Cycle Assessment (LCA) studies and value chain analyses of glass, as well as Environmental, Social, and Governance (ESG) risks within the value chain, are already well established, the maturity of sustainable oriented innovation (SOI) practices in the automotive glass industry remains insufficiently explored. A key gap identified in the literature is the absence of a comprehensive assessment of the automotive glass

industry, particularly regarding the integration of sustainability and innovation practices. Furthermore, there is a lack of coordination among existing studies and a limited number of actionable recommendations specifically tailored for OEMs.

The overall purpose of this study is to advance the level of sustainability at both Volvo Cars and within the automotive glass industry as a whole, focusing on the automotive glass component level. This includes consideration of both internal and external factors, such as suppliers and stakeholders, material flows, alternative materials, regulatory requirements, environmental impacts, and the strategic risks associated with automotive glass components.

The underlying purpose of this thesis is to investigate how sustainability within the automotive glass industry can be enhanced, while also increasing awareness of the environmental, social, and economic challenges faced by the industry [25]. Furthermore, the study aims to apply academic knowledge acquired throughout the project to examine SOI management strategies, with the intention of proposing improvements in the value chain, enhancing material circularity, reducing environmental impact, and mitigating supply chain risks. The following are the goals of the study:

1. **Subgoal 1**

   The first sub-goal focuses on evaluating and mapping the current value chain of automotive glass components, from raw material extraction to the retirement and end-of-life stages of the materials. This objective includes conducting an ESG evaluation analysis to identify the main sustainability-related challenges and opportunities associated with glass raw materials. It also involves assessing current material flows, CO_2 emissions and the use of recycled content. Under this objective, the first research question is formulated as follows:

   RQ1: How is the current automotive glass value chain configured and to which sustainability risks is it exposed?

2. **Subgoal 2**

   The second sub-goal is related to the evaluation of SOI and their application within the automotive glass industry. This includes examining innovation management literature on SOI practices in order

to evaluate its applicability in practice. The second research question is as follows:

RQ2: How are sustainability-oriented innovation practices applied in practice compared to their theoretical framework?

3. **Subgoal 3**

   The third sub-goal focuses on developing a recommendation roadmap for Volvo Cars and industry sector to enhance sustainability within in the automotive glass value chain. This objective includes assessing ongoing and emerging research trends, as well as the outlook of certification schemes within the automotive glass industry. Accordingly, the third research question is formulated as follows:

   RQ3: What actions can the automotive glass industry take to enhance its sustainability performance?

1.3 Delimitation

This study is primarily structured as a qualitative and strategic assessment, based on interviews conducted within the organization as well as with relevant project stakeholders. The project does not aim to further develop the technical aspects of the various solutions available on the market. Instead, it seeks to synthesize existing knowledge, evaluate alternative solutions, and support decision-making processes related to innovation and sustainability management at the component level.

Furthermore, the analysis is conducted using a single recent Volvo vehicle model, which serves as the reference throughout the project. Given the wide range of Volvo models, this approach facilitates data collection and subsequent analysis. The scope of the analysis is limited to float glass, specifically used in exterior glazing in components such as windshields, rear shields, roof and windows as it has been identified to represent a significantly larger volume compared to other types of automotive glass within the vehicle.

However, due to Volvo Cars' involvement in the project, confidentiality considerations must be respected. Consequently, certain information may be

redacted where necessary, and such omissions will be clearly indicated.

Chapter 2

Literature Review

The following chapter introduces the background theory on which the thesis is based.

2.1 Automotive Glass

The automotive glass sector is a mature and highly structured segment of the broader automotive industry, encompassing a wide range of applications and component types. Contrary to what the term "automotive glass" might initially suggest, glass is not limited to a few externally visible parts of the vehicle such as windshields, windows, and sunroof [26, 27]. This category extends beyond these elements and include additional components such as mirrors, sensors, and other auxiliary glazing elements, as well as specialized interior and display-related glass components [28, 29]. Furthermore, glass-related materials are also present in the form of advanced polymers reinforced with glass fibers, which enhance material strength, stiffness and thermal properties and performance characteristics [30, 31].

2.1.1 Glass as an engineering material

Glass materials share similarities with crystalline solids, as both are mechanically rigid, while also resembling liquids due to their disordered molecular structure. It is this dual character that contributes to their scientific interest, even among non-specialists [32]. From this perspective, glass is not classified as a ceramic, although it is often used in similar applications and exhibits properties comparable to those of ceramics. According to the

definition provided by ASTM C114 (Committee C28), this class of materials is characterized as highly engineered, high-performance, predominantly non-metallic, inorganic ceramic materials with specific functional attributes [33].

The mechanical and physical properties of glasses are largely determined by their composition; however, several common characteristics can be generally identified [34]:

1. Glasses are typically harder than metals.
2. Glasses exhibit tensile strengths in the range of 24–69 MPa.
3. Glasses are brittle materials with low ductility.
4. Glasses have a low coefficient of thermal expansion.
5. Glasses possess a low coefficient of thermal conductivity.
6. Glasses are effective electrical insulators.
7. Glasses are resistant to acids, solvents, chemicals, water, saline water, and alkaline solutions.
8. Certain types of glass can withstand high temperatures (e.g., soda-lime glass up to approximately 700 °C and fused quartz–silica up to approximately 1580 °C).

The relationship between oxygen and the cation in oxide compounds fundamentally influences the glass-forming ability of oxides. Four main types of oxides are typically used in glass fabrication, as shown in Table 2.1 [35].

Table 2.1: Types of oxides in glass fabrications

NO.	Oxide type	Characteristics	Examples
1	Main Glass Former	Suitable structures and low crystallization rates Form glass under slow cooling rates	SiO_2 B_2O_3, GeO_2, P_2O_5
2	Conditional glass formers oxides	Form glass under certain conditions	Al_2O_3; Bi_2O_3 WO_3; MoO_3
3	Intermediate oxides	Cannot form glass themselves but form glass in mixture with former oxides	TiO_2; ZnO; PbO; Zr_2O_3
4	Network modifier oxides	Cannot form glass themselves nether in mixture with former oxides They can modify the properties of glass by affecting Si-O bonds	MgO, CaO Na_2O K_2O

2.1.2 Type of glasses

Glass, as an engineering material, is widely used across various industries. Owing to its versatility, it can be classified in several ways: by application, by chemical composition, or by structural characteristics. Within the automotive industry, different types of glass are utilized, as illustrated in Figure 2.1. When classified according to application, automotive glass can be found in mirrors, lighting systems, displays, fillers, and other components [35].

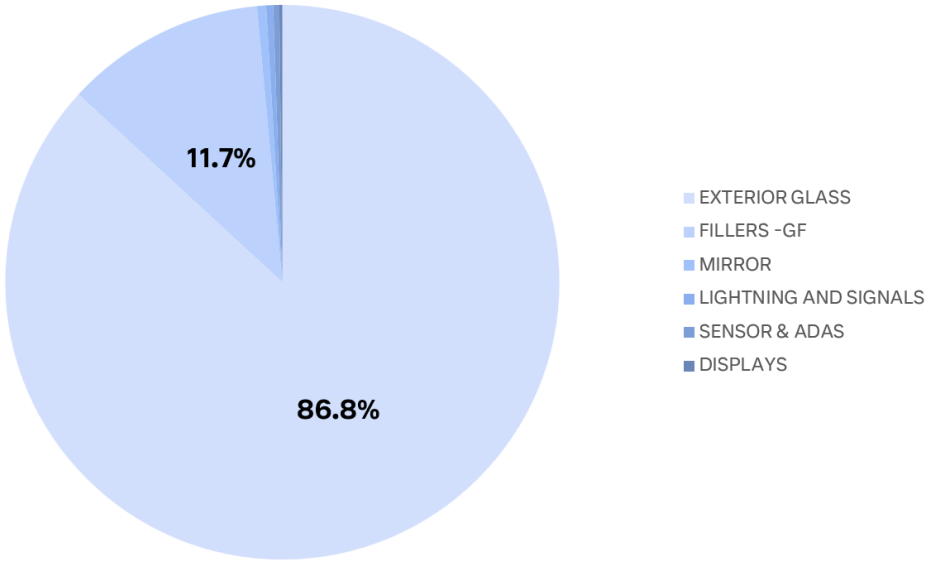


Figure 2.1: *Distribution of glass mass in a vehicle according to its application (diagram self-generated based on a Volvo vehicle material used).*

As outlined in the delimitation, for this study the focus is placed on exterior glazing due to its predominant presence in vehicles. This approach is justified by the significant concentration of mass, as exterior glazing accounts for approximately 87% of the total glass used in a vehicle. The exterior glass material can be classified according to the types of glazing found within the vehicle. The following types of glazing can be identified:

1. **Non-laminated Glass**

   Non-laminated glass consists of a single layer of glass. This type of glass, commonly referred to as tempered or toughened glass and is considered a safety glass. It is modified through thermal or chemical treatments to increase its toughness and enhance its resistance to frontal impacts. However, it remains susceptible to failure under lateral loading and impacts from sharp or pointed objects [36]. This is due to the internal tensions created during it manufacturing process as the core of the glass is subjected to compression stresses on its frontal surface, while on the corners and edges it is subjected to tension stress which make them vulnerable points fo the glass. This gives the glass high

impact enduring strengths [37].

This type of glass is characterized by its ability to fracture into small fragments when broken. This property is important for safety, as it reduces the risk of injury to passengers during an accident. Additionally, it offers performance advantages by allowing designers to use thinner and lighter glass while maintaining the required performance. Due to these characteristics, tempered glass is widely used in various automotive glazing applications and may also be incorporated into laminated glass systems [36].

The strengthening process is carried out prior to any lamination and can be performed using two main methods, thermal tempering (temperature shock) or chemical strengthening. Thermal tempering involves the heating of glass in a furnaces in rapid process followed by air quenching to generate compression stress at the surface level of the glass. While the chemical strengthening involves the use of salt-based materials to achieve similar result [38]. From a sustainability perspective, both methods present comparable environmental impacts. Thermal tempering requires higher energy input, whereas chemical strengthening involves the generation of molten salt waste, which can be recycled, although not with full efficiency. From a CO_2 emissions standpoint, chemical strengthening is generally considered more favorable than thermal tempering due to its lower energy requirements [38, 39].

2. **Laminated Glass**

   Laminated glass consists of multiple layers of float glass bonded together by a plastic interlayer, typically PVB. This type of glass is primarily used in automotive windshields due to its superior safety performance, although it can also be found in sunroofs, sidelites, and other glazing components within the vehicle. Its configuration provides high resistance to impact and penetration, as the plastic interlayer prevents external objects from entering the vehicle during a collision and helps retain the glass fragments adhered to the PVB layer [40, 41].

2.1.3 Raw materials

Automotive glass, as the name may suggest, is not solely composed of glass. While glass is the primary material, the component also incorporates various types of plastics and polymers to enhance its performance and facilitate its integration into the vehicle. Among the raw materials used in automotive glass, the fundamental ones include (Not including antennas or heating elements):

1. **Floated Glass**

   Float glass is the primary raw material used in automotive glass applications, undergoing several processing steps to meet the technical specifications of automotive components. It is most commonly produced from silica sand, soda ash, and limestone, although it may also incorporate recycled glass cullet to reduce CO_2 emissions and decrease the use of natural resources. A deeper analysis on materials materials, recycled content and materials flow can be seen in appendix A.

   As shown in Figure 2.2, the process begins with the mixing of raw materials into a batch, which is then fed into a furnace operating at temperatures of approximately 1550–1600 °C. At these temperatures, the materials undergo thermal softening and subsequent melting, forming a homogeneous molten glass in combination with the other constituents. Subsequently, the molten glass undergoes a refining and conditioning phase, during which it is cooled from approximately 1400 °C to 1100 °C before entering the float bath, where it acquires its flat geometry. The glass then passes through an annealing stage to relieve internal stresses, followed by quality control and cutting stages to ensure quality and required dimensions as final steps prior to delivery.

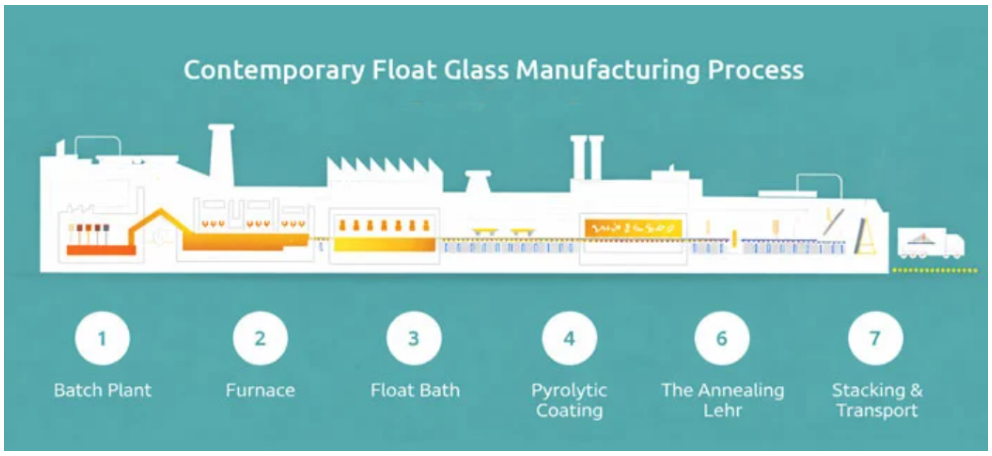


Figure 2.2: *Float Glass production diagram, figure elaborated by Saint Gobain Glass*

2. **Polyvinyl Butyral (PVB)**

   For certain types of automotive glass, PVB is used as an interlayer material. PVB is a high-performance thermoplastic resin that acts as an adhesive layer bonding two sheets of glass. This material is specifically engineered for such applications and exhibits properties that are particularly suitable for the automotive industry, including high optical clarity, strong adhesion, and high elasticity [42].

3. **Pins, seals, brackets, antennas and other components**

   In the automotive industry, basic glass components primarily consist of bent, tempered, and/or laminated glass. However, these components often incorporate additional elements such as pins, seals, brackets, or surface coatings, which enhance performance, enable proper installation, and provide additional functionality. In more advanced automotive glass applications, components may include complex features enabled by emerging technologies, such as integrated solar panels, switchable color-changing layers, light-reflective coatings, and other functional systems. Each of these features requires the use of additional materials beyond the basic glass composition [42].

2.1.4 Automotive glass component production process

Depending on the type of glass and its technical requirements, the production process may vary. However, as shown in Figure 2.3. The standard automotive glass manufacturing process typically begins with the cutting and polishing of float glass. The glass pieces then undergo painting and sintering to apply black bands and logos followed by the bending stage, where the glass is shaped according to the specifications of the respective vehicle model.

Following the bending stage, the glass may undergo a tempering process, if required, and subsequently lamination and autoclaving depending on the final product specifications. The components then undergo final inspection and dimensional verification before proceeding to the Added Value Operations (AVO) stage, where pins, seals, brackets, or additional coatings are installed or applied. Finally, upon completion of all processes, the glass is packaged and shipped to vehicle assembly facilities [42].

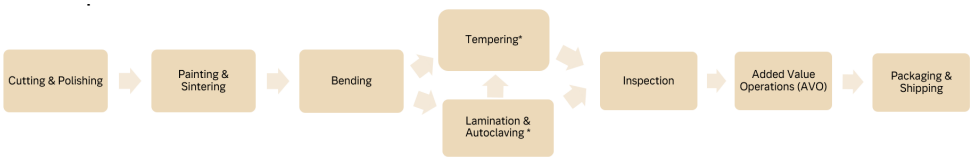


Figure 2.3: *Automotive glass production process (Diagram self-developed based on the Saint-Gobain Sekurit report). *The tempering and lamination & autoclaving process can be or not part of the automotive glass or be both together can be included depending on the design of the glazing.*

2.1.5 Life Cycle Assessments (LCA) analysis

The manufacturing of float glass is highly energy-intensive and, consequently, associated with significant CO_2 emissions. The industry-average global warming potential for flat glass production is approximately 1430 kg CO_2 eq., according to the National Glass Association (NGA), although this value may vary depending on the efficiency of the production process. [43].

Through a review of a Saint-Gobain Sekurit LCA report released in 2024, data on the manufacturing process and associated CO_2 emissions were collected per kilogram of laminated windshield glass produced in its

Poland facilities. The CO_2 emissions from the raw materials and float glass production process were calculated to be 1.27 kg CO_2 eq. per kg of glass. Emissions from the component manufacturing process were estimated at 2.03 kg CO_2 eq. Downstream emissions associated with the recycling stage were calculated to be 0.027 kg CO_2 eq. These LCA results indicate that the majority of CO_2 emissions occur during the component manufacturing stage [44] for the Poland Saint-Gobain windshield.

On the other hand, the same LCA analysis performed within the same organization but at facilities in different locations details that, for one kilogram of laminated windshield, emission values of 1.09 kg CO_2 eq. were generated from the production of raw materials and glass flotation, while the component manufacturing stage had a value of 0.548 kg CO_2 eq. This difference indicates that even within the same supplier, CO_2 emissions can vary depending on its sustainability strategies, sourcing and renewable energy use, as for this case that was identified as the main difference [45].

2.1.6 Automotive glass Value Chain and its sustainability challenges

To manufacture automotive glass, as previously described, different production processes are applied to flat glass. From a broader perspective of its value chain, from its origin to its end-of-life stage. It is important to note, that the preliminary value chain structure, used for the research of the list of sustainability issues, was developed using the standard Six Stages of a value chain as a base line as shown in Figure 2.4 [46, 47]. Through the adaption of the structure on the specific case of the automotive glass, the following 6 stages are identified:

1. Raw Material extraction: This stage consist represent the mining activities of silica sand, soda ash and limestone.
2. Glass Fabrication: It is the stage where the materials are used to produce the flat glass which is the main raw material of the automotive glass.
3. Component Manufacturing: It is the stage consisting of cutting, painting, shaping, tempering and lamination of the float glass to produce what is known as the automotive glazing.
4. Component Installation: Installation on the vehicle stage.

5. Use of the vehicle: Use of the glazing while the vehicle is in function. In this stage, it is possible that some users change some glazing due to breakage.

6. End-of-life: It is the stage when the vehicle reaches end of life and gets disposed with its glazing.

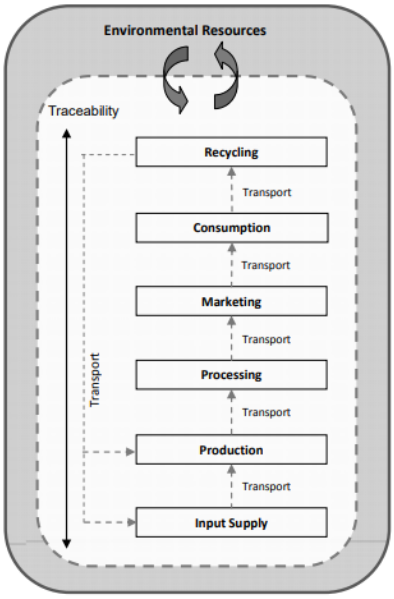


Figure 2.4: *Six stages of a value chain [46, 47].*

The automotive glass value chain is as detailed in the delimitation chapter focused only on the glass that can be found in the exterior glazing of the vehicle, this implies that Glass Fibers (GF), Polivinyl Butyral (PVB) (interlayer), plastic components, like pins, brackets and seals are out of the scope of the value chain. This decision was made, with the purpose of simplifying the analysis and elaborate a detailed value chain for the presented case. The automotive glass value chain diagram developed for the research is presented in Figure 2.5.

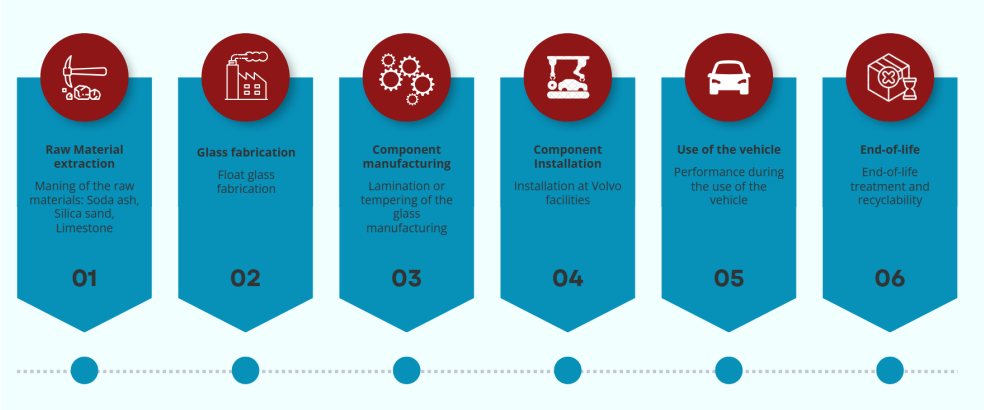


Figure 2.5: *Preliminary version of Automotive Glass Value Chain.*

Through a literature on the value chain presented, many challenges, as well as different sustainability issues were identified and the following is the preliminary list of problems identified at each stage are:

1. **Raw Material Extractions**

   (a) **Silicosis disease:** Silicosis is a respiratory disease caused by prolonged exposure to and inhalation of silica dust. Inhalation of silica particles leads to inflammation and irreversible scarring of lung tissue. Millions of individuals have been affected by this disease between 1990 and 2019 alone. In severe cases, silicosis can be fatal, while in less severe cases it may result in reduced lung function or long-term disability [48].

   (b) **Silica sand mafias**: The term "silica sand mafias" is commonly used to describe criminal activities involving the illegal extraction of silica sand from unauthorized locations. These activities result in significant social and environmental impacts, including land degradation and violations of human rights within affected communities. Such operations are most prevalent in countries including India, Indonesia, China, Bangladesh, Iran, and Oman; however, instances of illegal extraction have also been reported in Europe. These activities often involve the exploitation of local or migrant labor, and the extracted silica sand is typically used for domestic purposes, particularly in construction sectors driven by rapid urbanization [49].

   (c) **Synthetic Soda ash production environmental impact**: Synthetic soda ash production via the solvay process is associated with

high CO_2 emissions, estimated at approximately 1.2 tCO_2 per ton of produced soda ash. In comparison, natural soda ash production generates significantly lower emissions, approximately two to four times less, corresponding to around 0.3-0.5 tCO_2 per ton [50]. These elevated emissions primarily result from the calcination of limestone and the high energy intensity of the process, which often relies on coal or other fossil fuels [50]. Additionally, the process produces by-products such as calcium chloride and ammonia, which can contribute to water and soil pollution if not properly managed [51].

(d) **Silica Sand mining impact on fauna and flora**: The mining of silica sand has significant impacts on fauna and flora due to the extensive disruption of land and habitats caused by extraction activities, which involve the removal and displacement of large volumes of sediment. This process can result in complete habitat loss at ground level [52]. Additionally, global annual consumption of sand and silicates is estimated to be approximately 70,000 times greater than the combined volumes of cobalt, lithium, gold, tantalum, tin, and tungsten, highlighting the scale and intensity of its extraction [53].

2. **Glass flotation & Component manufacturing**

(a) **Energy-Intensive Process**: Glass production is inherently energy-intensive, as the primary raw material, silica sand, requires melting temperatures of approximately 1500 °C. Subsequent stages, including floatation, shaping, and lamination, also demand substantial energy input. Although suppliers are increasingly pursuing decarbonization strategies within their operations, the industry remains largely dependent on natural gas as a primary energy source. It is estimated that the flat glass industry consumes approximately 61,000 TJ of energy in European facilities alone, corresponding to around 0.4% of total European energy usage [54].

(b) **Risk of PFAS**: In conventional glass applications, per- and polyfluoroalkyl substances (PFAS) are typically not present. However, in the case of hydrophobic glass, which incorporates external coatings, PFAS may be used as part of the material composition. Additionally, in applications where glass integrates

photovoltaic technologies, PFAS are often applied as protective layers for the cells to prevent moisture-related degradation. In the automotive sector, hydrophobic coatings are commonly applied to windshields, mirrors, and camera sensors to enhance visibility and performance [55]. This raises concerns from a regulatory perspective, as the European Union REACH framework is progressively introducing stricter restrictions on the use of PFAS due to their environmental persistence and potential health risks [56].

3. **Use in the Vehicle & End of life**

   (a) **Automotive Aftermarket:** In 2023, more than 41 million windshield replacements were recorded globally due to accidents, wear, and weather-related damage. Additionally, the expansion of mobile automotive glass replacement services has reduced average service times by approximately 34% in urban areas. The automotive glass aftermarket constitutes a significant source of waste, as approximately 24% of global demand for automotive glass is attributed to this segment [57].

   (b) **Low level of circularity:** According to a LCA conducted in 2023 by Saint-Gobain Sekurit at one of its facilities in Poland, only approximately 1% of end-of-life recycled cullet is reintegrated into production, while the remainder is either disposed of in landfills or downcycled into applications such as glass fibers [44]. Other competitors have been identified to achieve slightly higher rates, around 8%; however, these levels remain relatively low, particularly considering that glass can theoretically be recycled indefinitely without degradation in quality [43]. Through interviews conducted as part of this study, it was determined that these low recycling rates are primarily due to contamination issues that affect the recyclability of the material, as well as limited economic incentives, since recycling is often not considered sufficiently profitable.

   (c) **Automotive glass quality requirements:** To ensure compliance as a responsible automotive glass product, it must adhere to different homologation frameworks, such as ANSI, ECE, and CCC, which makes the implementation of design changes more challenging. These standards impose stringent technical

requirements, particularly for windshields. This is due to the critical role of the windshield as the interface between the driver and the external environment, requiring high levels of optical clarity, impact resistance, and overall performance [58, 59, 60].

As shown, the automotive glass value chain presents several challenges throughout each of its stages, representing a significant impact on the industry and highlighting the importance of further investigation of this material.

2.2 Sustainable sourcing

In the automotive industry, OEMs selectively choose build-to-print components from supplier-sourced components or systems when it is strategically beneficial. This choice is based on the importance of components to the core strategy of the product. Certain areas of the vehicle are considered standardized within the industry and are therefore less critical for differentiation. In contrast, other components are identified as key elements for achieving competitive advantage [61].

Components supplied by external suppliers and subsequently integrated into vehicles require effective supplier management and a well-defined outsourcing strategy. This strategy represents a critical aspect of value chain management, which has evolved significantly over time. One widely adopted concept in this context is Green Supply Chain Management (GSCM), which refers to the integration of sustainable practices into supply chain management [62].

2.2.1 Green Supply Chain Management (GSCM)

GSCM involves evaluating the sustainability performance of all processes within the supply chain, from sourcing and production to distribution and final disposal [63]. However, it is not limited to these stages; it also entails a broader responsibility extending beyond direct suppliers to include sub-suppliers across the entire value chain as seen in Figure 2.6. Furthermore, GSCM influences decisions related to product design, manufacturing processes, and end-of-life management strategies, all of which can significantly impact the overall sustainability of the supply chain [64].



Figure 2.6: *Procedure or Steps of Green Supply Chain Management from Saada [65] and elaborated by Hariyani [66]*

This concept is driven by several factors, including cost reduction, improvement of brand image, emerging regulations, international standards, and competitive pressures [67]. The implementation of a GSCM strategy can provide multiple benefits, such as waste reduction, a lower carbon footprint, improved resource efficiency, and enhanced environmental performance [68]. However, these benefits do not depend solely on the focal organization, but rather on the performance of the entire value chain. To achieve these outcomes, various strategies can be adopted, such as closed-loop systems that support circular economy principles, and sustainable sourcing practices that prioritize environmentally and socially responsible materials and suppliers [69].

A practical example of the implementation of a GSCM strategy has been documented in Chinese firms. These firms adopted internal environmental management systems, redesigned products to enhance sustainability, and developed waste recovery systems. Such implementations contributed to improvements in environmental performance, operational efficiency, and long-term economic benefits, despite requiring significant short-term investments [64].

2.2.2 Sustainable sourcing strategies

Within sustainable sourcing strategies, several key concepts can be identified, such as increasing the use of recycled content or adopting more environmentally friendly materials, including biodegradable alternatives, to minimize environmental impacts [63, 67, 68]. Evaluating suppliers based on their sustainability performance is no longer merely a “good-to-have” parameter, but a fundamental requirement to remain competitive in the market, not only due to brand perception and sales performance, but also to comply within regulations and increasing governmental pressure [70]. In the automotive sector, this is enforced by European Union regulations on circularity in vehicle design and the management of end-of-life vehicles [71].

As previously described, sustainable sourcing involves incorporating

additional criteria into supplier evaluation, while maintaining a balance with traditional decision factors such as cost, delivery time, production capacity, and quality performance. Achieving this balance presents a significant challenge, often requiring organizations to adopt different frameworks and involve multiple decision-makers to achieve their objectives [72]. Depending on the different strategies applied, Akhavan et al. (2017) argue that five archetypes can be identified [73]:

1. **Minimalist:**
   This category includes companies that implement only a limited number of sustainable sourcing practices, and typically at a minimal level. These firms focus on isolated initiatives and remain at an early, exploratory stage [74]. This is often due to their limited interaction with end consumers and the relatively low level of feedback pressure they experience [75]. Such companies are generally oriented toward business-to-business (B2B) markets, such as mining firms or natural resource manufacturers [76]. Although direct consumer pressure is limited, these firms are increasingly subject to sustainability related demands from their immediate clients. As a result, this archetype is becoming less prevalent, particularly among organizations aiming to remain competitive in the market [76].

2. **Social risk manager:**
   This archetype is characterized by a selective focus on mitigating social risks, such as child labor and occupational health and safety [74, 77]. Typically, supplier agreements include a code of conduct that must be adhered to, supported by monitoring systems designed to ensure compliance [78].

3. **Collaborator:**
   This archetype is characterized by relatively high levels of internal integration and active participation in industry-wide sustainability initiatives [74]. However, the degree of practical implementation remains comparatively limited. These companies are primarily distinguished by their involvement in collaborative associations that address sustainability challenges at the industry level. Examples include firms participating in initiatives targeting global issues such as sustainable palm oil production, involving companies like Unilever and

Migros, as well as associations addressing conflict minerals, such as IRMA, which includes technology companies such as Apple, Tesla, and Intel [79].

4. **Compliance networker:**
   This archetype is characterized by a high level of activity, relying on standards and certifications to ensure compliance with social and environmental practices [80]. These companies actively engage in collective initiatives and utilize them to enhance legitimacy while reducing transaction costs [81].

5. **Proactivator:**
   This archetype is characterized by a strategic and proactive approach, with a strong focus on supplier development to achieve competitive advantage [80, 82]. Proactive organizations engage directly with their suppliers rather than relying solely on collective initiatives. For instance, some companies, particularly in mass production industries, implement strategies aimed at enhancing supplier sustainability by offering increased demand and additional business opportunities. These organizations may also develop dedicated platforms and tools to support and accelerate improvements in suppliers' sustainability performance. A notable example is H&M, which, through the H&M Foundation, supports the textile industry in reducing greenhouse gas emissions and promoting a sustainable transition, with a strong emphasis on both social and environmental dimensions [83].

It is important to highlight that due to the complexity of different industries, companies may apply different strategies to different types of suppliers. In practice, firms do not strictly fit into a single archetype, as market behaviors and patterns require organizational adaptability [74]. An interesting case is Patagonia, which implemented changes in both the design and concept of its products to ensure a sustainable value chain and responsible sourcing. The company began using only organic cotton and recycled materials to mitigate environmental impact and introduced programs such as Worn Wear to support customers by repairing garments and extending product life cycles. In addition, Patagonia engages extensively in environmental activism and carbon compensation, investing in renewable energy and community projects [84]. This demonstrates that while the company may not directly solve global sustainability challenges, it implements significant internal changes and

proactive initiatives, meaning it does not fit neatly into a single archetype.

On the other hand, companies that operate with greater caution and stricter technical performance requirements, such as those in engineering-intensive sectors, tend to adopt different approaches. In many cases, they focus on material-specific strategies for resources such as aluminum, copper, cobalt, and other scarce or critical materials with distinct market dynamics. In this context, and with the objective of conducting responsible business, the automotive industry seeks to improve conditions related to each material of concern while simultaneously balancing other requirements, such as technical performance, value chain resilience, legal context, and innovation levels, in order to remain competitive. An example of this is Volvo Cars and its blockchain implementation for cobalt, lithium and nickel, a system designed to track materials from the mine to the final product. Since cobalt is highly associated with child labor, this initiative highlights the effort and type of implementations being adopted in such value chains [85].

2.3 Sustainable oriented innovation (SOI)

Interest in the development of SOI has steadily increased in recent years, driven by growing environmental, social, and economic concerns. Consequently, an increasing number of companies are motivated to innovate in order to achieve their long-term visions and strategic objectives [86]. Large and well-established companies tend to become less innovative over time as their core abilities slowly convert into core rigidities [87]. Core capabilities are what makes a company stand out from the rest, however such capabilities are at times deeply rooted within company value and create an internal resistance against innovation when newer technologies, or newer customer behaviour comes along [88].

2.3.1 SOI objectives

The main objectives of SOI are defined as transforming organizational values to generate sustainable economic, environmental, and social outcomes, rather than focusing solely on profit maximization [86, 89]. Harsanto et al. (2022) note that social enterprises pursue objectives that extend beyond maximizing profits to include the reduction of negative environmental and social impacts.

This is because such organizations embed sustainability within the core of their operations, striving to minimize harm rather than prioritizing profit alone [90]. Similarly, the primary objective of SOI is to be embedded within the core innovation activities of firms and implemented at the strategic level of the organization [89].

2.3.2 SOI benefits

The benefits of SOI include the sustainable development of firms [86, 89, 90]. SOI supports organizations in achieving more responsible and sustainable performance while maintaining profitability as a key objective [86, 90]. Firms can enhance their economic performance by actively reducing resource consumption, waste generation, and emissions [86, 90]. At the same time, SOI contributes to social outcomes by improving well-being, promoting equity, and strengthening stakeholder engagement. Furthermore, it supports economic performance by enabling the development of long-term competitive advantages and innovation capabilities [86, 90].

2.3.3 SOI practices

Sustainable development is widely defined as development that meets the needs of the present without compromising the ability of future generations to meet their own needs [91]. Achieving sustainable development requires a societal alignment that balances social, economic, and ecological objectives [92]. Cabezas (2003) conceptualizes innovation as an integrated human natural system in which social, environmental, and economic factors play a critical role in overall system functionality [93].

As emphasized by Vollenbroek (2002), truly sustainable innovation requires more than incremental improvements, involving significant organizational change and enhanced cross-sector collaboration to enable the transition toward sustainability oriented solutions [92]. A lack of cross-functional or cross-divisional coordination tends to reinforce exploitative organizational behaviors rather than exploratory ones. Such continuous focus on exploitation may lead organizations to improve existing processes without addressing fundamental challenges, effectively resulting in “doing the wrong things better” [24].

Vollenbroek (2002) further emphasizes that sustainability and innovation are closely interconnected as drivers of long-term competitive advantage for organizations [92]. Similarly, Arekrans et al. (2023) highlight that sustainability initiatives are strongly linked to long-term competitiveness [24]. It is therefore recommended to maintain a balance between exploration and exploitation in order to achieve meaningful and sustainable long-term outcomes [24]. However, organizations are often more incentivized to prioritize exploitation, as it provides more certain outcomes and involves lower short-term risks. Over-reliance on this approach may ultimately lead to long-term decline if exploratory activities are insufficiently supported [94]. Yunfei et al. (2024), further support these findings, demonstrating through their analysis of business analytics that enhancing organizational ambidexterity is positively correlated with improved sustainability outcomes [95]. Similarly, a quantitative study of 150 mid-sized German engineering firms finds that increasing the focus on exploratory activities contributes to enhanced competitive advantage [96].

Therefore, March (1991) and Alänge et al. (2018) recommend that established organizations are always required to allocate resources into both exploitation of old certainties and exploration of new possibilities to maintain or to gain competitive advantage over competitors [87, 94]. Ambidexterity definition is "the ability to pursue different objectives simultaneously", in organizational terms it is most often used to refer to the strategy used for balancing between exploitation and exploration [87, 97]. Through a review on modern organizational ambidexterity theory, four types are identified, namely structural, managerial, sequential and contextual [98, 99].

Through a systematic review of 58 academic and practitioner articles done by Stoiber et al. (2023) on radical disruptive innovation, they were able to produce a theoretical framework that synthesises ambidexterity types, however the authors mention that there is no single best method for balancing exploitation and exploration, however based on barriers organizations face such as culture inertia that favours routine works or risk aversion the ambidexterity type could differ [100]. Arekrans et al. (2023), note that most commonly implemented sustainability actions tend to focus on the exploitation of existing resources. As a result, they argue for a stronger emphasis on radical thinking and exploratory approaches, suggesting that developing entirely new sustainable business models for waste management may be more effective for

achieving long-term impact than merely improving existing recycling methods [24].

As highlighted by Adams et al. (2016), it is important for SOI activities to be approached from a systems-level perspective [86]. SOI must therefore be embedded at the business level, requiring new ways of thinking in how organizations operate and create value [101]. Circular economy represents an approach aimed at moving away from the traditional linear "take–make–waste" model toward a closed-loop system, where waste is continuously reintegrated and minimized [102]. Circular economy principles focus on narrowing, slowing, and closing resource flows by retaining value from products, materials, and activities over time [101]. In this context, circular economy provides a strong foundation for the implementation of SOI, as it promotes continuous processes of efficient and effective resource utilization within innovation activities [101].

While SOI may appear to be an inherently beneficial approach, its implementation faces several challenges. When considering the circular economy, it is often primarily associated with recycling and recovery processes. However, key stages such as manufacturing, assembly, and product design are frequently overlooked, despite their critical role in enabling circular outcomes [103]. Another significant challenge relates to material selection, particularly regarding material purity and ease of recovery. Identifying materials that facilitate circularity while simultaneously meeting stringent quality and performance requirements remains a complex task. In more complex systems, these challenges are further amplified, making the implementation of circular economy principles increasingly difficult [103]. Kalmykova et al. (2018) recommends placing greater emphasis on upstream actions when developing solutions, as these stages can be more directly influenced by firms and have a significant impact on overall circular resource flows [103].

2.3.4 SOI maturity

Through the foundational study by Adams et al. (2016), which analyzed the environmental management strategies of 50 firms, three levels of maturity in SOI were identified: reactive (At the reactive level, sustainability actions are limited and mainly respond to external pressure), embedding (Sustainability

is integrated into organizational processes), and transformative (Sustainability shapes the organization's purpose, strategy, and value creation), as illustrated in Figure 2.7 [86]. The assessment of organizational maturity is based on three key contextual dimensions:

1. The first dimension is the scope of innovation, which can be categorized as technical or people-oriented. Technical innovation tends to focus primarily on environmental challenges, whereas a people-oriented scope addresses more complex socio-technical issues, including regulations, usage practices, and cultural considerations [86].

2. The second dimension concerns the integration of SOI within the organization, distinguishing between stand-alone and integrated approaches. This dimension reflects the degree of organizational alignment in implementing SOI strategies, indicating whether such practices are confined to individual departments or embedded across the entire organization [86].

3. The third dimension relates to the implications of SOI, distinguishing between insular and systemic outcomes. This refers to whether SOI efforts are directed toward internal improvements within the organization or aimed at generating broader impacts on socio-economic systems [86].

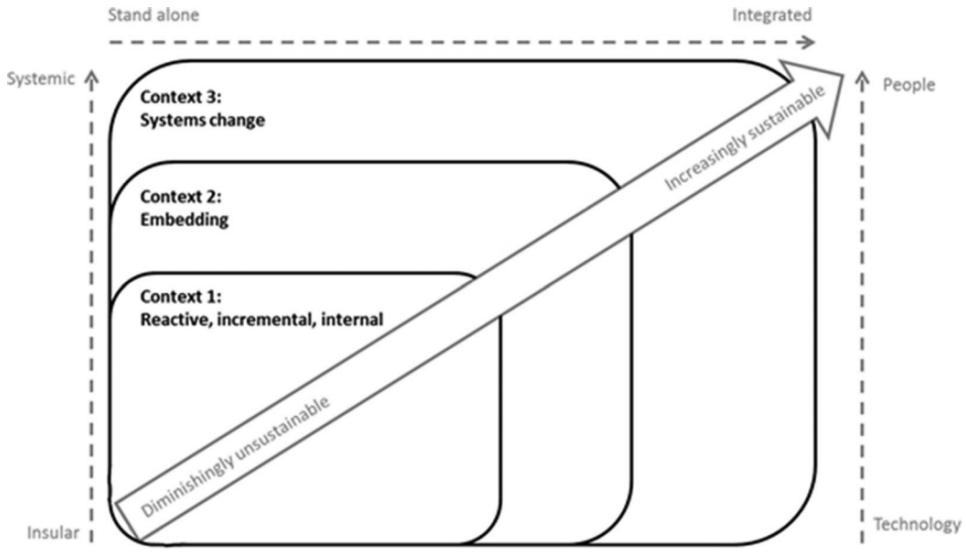


Figure 2.7: *Assessment model for level of SOI maturity [86]*

Chapter 3

Research Methodology

The purpose of this chapter is to provide an overview of the research process and the methods applied in the thesis project.

3.1 Research Process

The research project followed a series of sequential steps to address the research questions and meet the objectives established in the previous chapters. It is important to note that the project approached the same problem from two complementary perspectives. On the one hand, the sustainability challenges inherent to automotive glass were evaluated by identifying the societal, environmental, and business risks present throughout its value chain [104, 105]. On the other hand, the project assessed the maturity of SOI practices in the sector by identifying key drivers and barriers and examining how the organization responds to these conditions [86, 106]. This second perspective is critical for advancing the organization's sustainability performance not only in relation to automotive glass, but also across broader sustainability initiatives. Under these two perspectives, each stage of the project included research and analysis activities designed to further explore both dimensions. The project consisted of five major stages as illustrated in Figure 3.1.

The first stage focused on problem identification. In line with the project's two perspectives, this stage first involved reviewing both academic literature and market-based sources related to the automotive glass industry in order to establish an initial understanding of the value chain and its main challenges [107]. In addition, to address the second perspective and to refine the value

chain assessment, this stage included the development of interview questions and the subsequent coding of the collected interview data to capture the maturity level of innovation practices among the relevant stakeholders [108]. The aim here is to analyze additional value chain risks not identified in literature

The second stage of the project consisted of problem analysis. In this stage, different methods were applied to analyze the data collected for both project perspectives, namely the automotive glass value chain and SOI practices within the automotive glass industry. The objective of this stage was to identify the key issues and rank them according to their priority [109], as well as to identify the SOI dynamics that are present in Volvo's environment.

The third stage involved mapping potential recommendations and solutions within the project. During this phase, different solutions addressing the key problems identified in the previous stage were explored. With the support of stakeholders, a feasibility (materialization) analysis of the proposed solutions was conducted, and a preliminary implementation road map was developed [110].

Finally, the fourth stage focused on validating and refining the preliminary road map. The objective is to present the final outcome and communicate it to all involved parties. Through these activities, feedback was collected and used to further improve and finalize the road map.

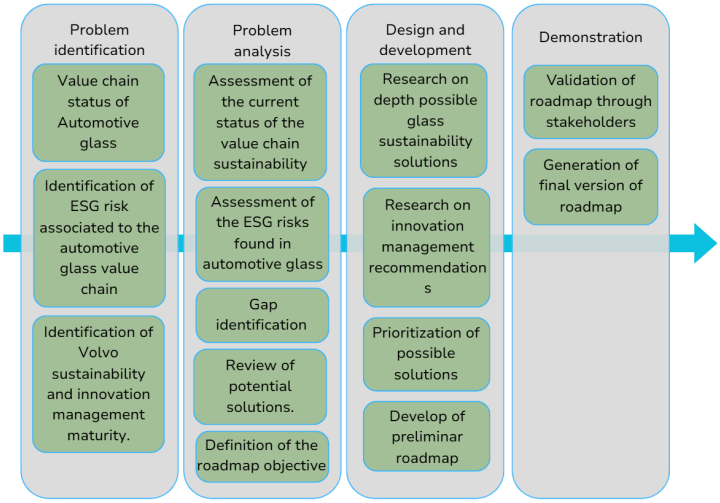


Figure 3.1: *Research Design Diagram*

As detailed in the Research Gap, the project aims to conduct research from two perspectives. Its objective is to analyze the value chain by identifying the high-priority ESG risks and, additionally, to identify the SOI dynamics that are shaping this industry sector. Based on this analysis, the project seeks to develop recommendations on how to improve the sustainability level of automotive glass from both a sustainability and a business perspective. Under this approach, the project followed the framework detailed in the following Figure 3.2:

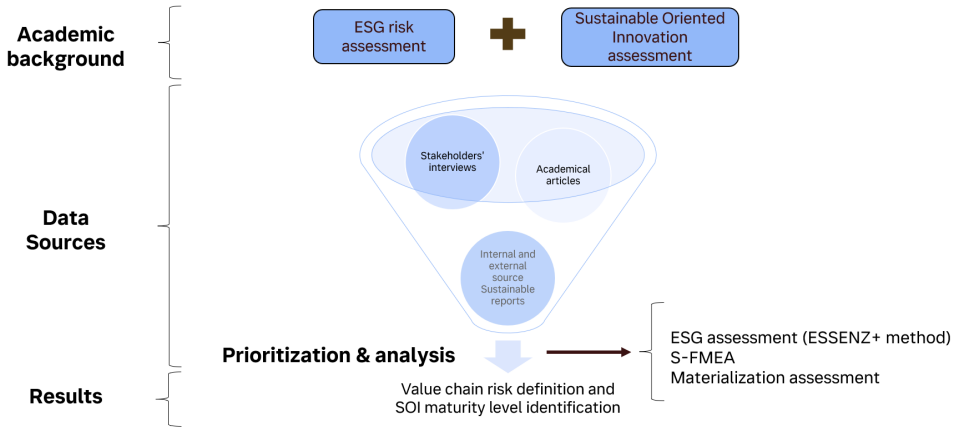


Figure 3.2: Framework followed for the research project

3.2 Data Collection Methods

For the development of the project, multiple sources and data collection methods were used. Throughout the study, both primary and secondary data were required. Primary data were collected through semi-structured interviews and feedback from relevant stakeholders involved in the project. The resulting material was subsequently analyzed using thematic analysis and coded accordingly [111]. Secondary data were collected from internal and external organizational websites and reports, as well as through a literature review of academic articles.

3.2.1 Primary data source: Semi-Structured Interviews

The primary data source was obtained through semi-structured interviews. These interviews were structured into two main sections. The first section followed the guide specified in Appendix C and was conducted uniformly across all stakeholders. The second section consisted of stakeholder-specific questions designed to further explore sustainability risks and to identify additional issues not currently addressed in the literature.

The interviews were conducted with both internal and external stakeholders identified throughout the study. These stakeholders were selected based on their direct involvement in the project and their relevance to the research objectives. Their perspectives, reflecting a range of roles and responsibilities, were essential for ensuring the reliability and validity of the findings. The stakeholder mapping is presented in Appendix D, and Table 3.1 provides an overview of the interview dates for each participant, as well as a stakeholder code used for tracking comments in subsequent analysis.

Table 3.1: Interview plan and dates

Stakeholder	Stakeholder Code	Date
Design representative	*S1*	16 March 2026
Manufacturing specialist	*S2*	6 March 2026
Legal analyst	*S3*	4 February 2026
Sustainability representative	*S4*	25 March 2026
Circularity specialist	*S5*	4 April 2026
Buyer specialist	*S6*	4 April 2026
Sustainable third party scheme 1	*S7*	9 March 2026
Sustainable third party scheme 2	*S8*	20 Feb 2026
Supplier representative	*S9*	24 March 2026
Recycler representative	*S10*	10 April 2026

3.2.2 Secondary data source: Literature Review

As part of the secondary data collection, various sources of information were reviewed and compiled. This information is classified according to the type of source. All the conducted research was synthesized and presented in Chapter 2, which provided foundational knowledge upon which this project is built.

1. **Academic literature Review**

   Different academic sources were reviewed and evaluated. The topics and search keywords used are detailed in Table 3.2. The reserch words were used in research engines, ResearchGate, Scopus and KTH library primo to find relevant litterature for the study.

2. **Business Websites and Sustainability Reports**

   In order to develop the value chain, business websites and sustainable reports were reviewed for different suppliers. This information was contrasted between them and evaluated according to the current situation of Volvo Cars. Due to the confidentiality agreement, the details of this data source is not included in this report.

3. **Internal organizational documentation review**

Table 3.2: Literature review fields

Topic	Research words
Automotive glass value chain mapping	'Silica sand mining', 'Glass production processes', 'Soda ash production', 'After-market volume and sales' . 'Automotive Glass recycling process'
Sustainable oriented innovation practices	'Sustainable oriented innovation', 'Sustainable business', 'Circular economy', 'Sustainable oriented innovation practices'
Sustainable sourcing	'Sustainable sourcing','Green Supply Chain Management', 'Sustainable sourcing strategies'

Inside Volvo Cars internal practices and guidelines, the organization already has some strategies placed and detailed, like, compromises and objectives related to sustainability, specific goals and objectives per material, list of materials of concern and its management procedures, codes of conduct for suppliers and methodologies standardized through the entire organization. This list of strategies and practices are detailed in their internal website source, and these were part of the data gathering of the research project.

3.3 Data Processing and Analysis Methods

Through all the stages of the research project, different data processing and analysis methods were used:

3.3.1 Problem identification & Analysis stages

To identify and define the challenges within the automotive glass value chain, considering both sustainability perspectives and innovation management practices, gathering of data as detailed in Subchapter 3.2 was made, followed by an initial ESG analysis showcased in the literature review chapter. This approach enabled a deeper understanding of the findings, supported the refinement of the study, and facilitated the identification of specific

sustainability and innovation practices within the automotive glass sector.

3.3.1.1 ESG risks assessment: classification and validation

An ESG assessment refers to the evaluation of Environmental (E), Social (S), and Governance (G) risks that an organization may face. Various frameworks exist to identify and classify these risks, with multiple dimensions considered within each of the three pillars [112, 113]. One of the commonly applied frameworks is the ESSENZ+ method. The method asses following:

1. **Social & Governance**
    (a) *Social risk:* This aspect considers various categories, including child labor, forced labor, discrimination, fair wages and living wages, cases of abuse, labor rights, and health and safety conditions [114].
    (b) *Governance risk:* This includes categories such as corruption and corporate governance practices.
2. **Environment**
    (a) *Environmental risk:* This dimension considers potential impacts on environmental aspects such as protected or conserved areas, biodiversity, deforestation, water stress, and environmental policies.
    (b) *Environmental impact:* This sub-dimension aims to quantify the scale of impacts, including categories such as damage to human health and ecosystems [115].
3. **Supply risk:** This dimension evaluates aspects such as production concentration, co-production dependencies, and trade-related risks.

Figure 3.3 summarizes the categories included within each dimension and sub-dimension of the ESG analysis. This list is based on the categories defined in the ESSENZ+ method [116, 117]. However, it does not include the full set of categories proposed by the ESSENZ+ framework. This selection was made to ensure alignment with Volvo Cars' internal methodologies.



Figure 3.3: *Dimensions, sub-dimensions of the ESSENZ+ method for ESG analysis. For this version of the method, some categories were omitted.*

It is important to highlight that a preliminary ESG analysis was presented in the literature review chapter, where a set of issues was identified for each stage of the production process. To ensure alignment with the ESSENZ+ framework, additional research was conducted using various sources, including material risk lists by country, as well as rankings related to fair wages and working conditions [118], among others. Some of the important sources used to complement were:

1. **Social & Governance:** The International Labour Organization conventions per country, as documented in NORMLEX, were considered [119]. Additionally, United Nations country reports and the 2024 List of Goods Produced by Child Labor or Forced Labor published by the U.S. Department of Labor were used [120].

2. **Environment:** Information was gathered from literature reviews and sustainability reports of key companies across the different stages of the value chain, as well as from academic research focusing on sustainability assessment and recommendations for silica sand and soda ash mining [121, 122].

3. **Supply risk:** Data sources included statistical platforms for market distribution and identification of key industry players, reports from the United States Geological Survey (USGS) [123], and analyses and reports published by the European Union [124].

Another source of information for the ESG risk assessment was the stakeholder interviews, specifically the second block of interview questions

as seen in Appendix C. Through these interviews, additional issues within the value chain were identified and subsequently classified according to the ESSENZ+ framework. These findings were then incorporated into the final version of the value chain diagram and the corresponding list of ESG risks.

3.3.1.2 Interview Data Encoding

As detailed in the data gathering subchapter, the first block of questions consisted of a semi-structured interview. This qualitative approach was used to examine how SOI is practiced within the automotive glass industry and to identify the rigidities and constraints that shape these practices [94, 125].

The data were collected following a predefined interview guide to ensure consistency across respondents, while allowing flexibility for elaboration [126]. The empirical data were analyzed using thematic network analysis, following the framework proposed by Attride-Stirling (2001) [127]. The analysis consisted of three stages:

1. **Text Reduction (Stage A):** Interview transcripts were coded based on recurring issues related to organizational rigidities, collaboration, ambidexterity, and sustainability focus. These codes were subsequently grouped into basic themes, organizing themes, and global themes [127].

2. **Exploration of Themes (Stage B):** The relationships between the identified themes were examined to understand how organizational structures, decision-making processes, and innovation practices influence sustainability outcomes [127].

3. **Integration of Findings (Stage C):** The final stage focused on interpreting the global themes in order to draw conclusions regarding how SOI and ambidexterity are implemented in practice [127].

This method enabled a structured yet flexible analysis of complex organizational and sustainability-related dynamics. For data processing, a tool called Granola was used. Granola is an AI-based software program primarily designed for note-taking and the transcription of meetings and interviews [128]. The tool was utilized to support note-taking during the conducted interviews. Its use was always disclosed prior to each interview and applied only with the consent and permission of the participants [129].

3.3.1.3 Sustainable FMEA (S-FMEA)

In order to determine which risks to prioritize, an S-FMEA (Sustainability Failure Mode and Effects Analysis) approach was applied for ranking and prioritization. FMEA is a widely recognized methodology used for failure mode analysis and risk assessment, originally developed in the aerospace industry [130], and later adopted across various sectors, including automotive, nuclear, and mechanical industries. Its primary objective is to identify potential failure modes and evaluate their effects [131]. These potential risks are assessed using a Risk Priority Number (RPN), which supports the identification of the most critical issues. In traditional FMEA applications, each failure mode or risk is evaluated based on three key criteria:

1. Severity: Measures the impact level of the risk into the final user, direct client and organization itself.
2. Occurrence: Evaluates the likelihood of the issue to happen.
3. Detection: Assess the difficulty to detect the problem.

These three parameters seek to evaluate all the layers of the risk and adapt it to the current situation of the organization. But, for the case of sustainability and value chain assessment, the tool requires some modifications and integration with other tools in order to properly comply with the objective [132].

For this case study, a sustainability-oriented FMEA (S-FMEA) was applied, which represents an adaptation of the traditional FMEA method to address value chain risks. In this context, failure modes are replaced by the sustainability-related risks identified across the value chain [133]. Regarding the effects assessed, they are subsequently evaluated based on their severity according to updated sustainability-oriented parameters:

1. **Local value chain effect:** This category includes the effects occurring at the specific stage of the value chain where the risk arises.
2. **System/Business effect:** This category encompasses all effects of the risk related to business and system performance.
   (a) *Supply chain effect:* Effects that impact the supply chain, including aspects such as material sourcing, logistics, energy supply, and related activities.

(b) *Business/Financial effect:* Effects related to costs, pricing structures, and business strategies that may be influenced by the risk.

(c) *Compliance & Legal effect:* Effects that directly affect compliance with regulations and legal requirements.

3. **End effect:** This category evaluates the broader impacts of the risk on the organization (Volvo Cars), end users, as well as society and the environment.

   (a) *OEM effect:* Assesses the impact on Volvo Cars, including business performance, brand reputation, and potential organizational changes.

   (b) *User effect:* Evaluates how the end user may be affected by the risk, including aspects such as brand perception, health, and safety (e.g., exposure risks).

   (c) *Societal/Environmental effect:* Considers broader impacts on society and the environment, including CO_2 emissions, biodiversity loss, and other environmental consequences.

An additional modification to the FMEA methodology was the exclusion of the Detection parameter from the RPN. This decision was based on several considerations. First, it enhanced the practicality and applicability of the analysis. Second, due to the nature of the identified risks which are inherent to the value chain and characterized by broader systemic impacts detection was considered more appropriately addressed through preventive and systematic approaches rather than corrective measures [134].

Finally, this adjustment aligns with Volvo Cars' internal standards as defined in the VTDS manual and associated guidelines. Under this approach, risks were assessed based on their level of criticality, determined by the combination of severity and occurrence values, following the matrix presented in Figure 3.4 [135]. Based on these modifications, the S-FMEA template used for the analysis is presented in Figure 3.5.

$$\text{Criticality} = \mathit{Severity} * \mathit{Ocurrence}$$

Criticality Matrix						
Probability	5	10	30	100	100	300
	4	10	30	75	100	300
	3	10	30	75	100	300
	2	10	30	50	75	300
	1	10	30	50	75	300
		1	2	3	4	5
		Severity				

Figure 3.4: *Criticality matrix, self-elaborated based on Volvo Cars standard*

List of Risks		Risk Effect														
			System / Business Effect			End Effect										
Number of risk	Risk Definition	Local Value-Chain Effect	Supply Chain Effect	Business / Finacial Effect	Compliance & Legal Effect	OEM Effect (Volvo)	User Effect	Societal / Environmental Effect	Severity	Motivation	Occurance	Motivation	Criticality	Index	Countermeasures	Details

Figure 3.5: *S-FMEA Template used for value chain risk analysis*

3.3.2 Design & development stage

For the purpose of designing and presenting a set of recommendations to improve the current sustainability performance of the automotive glass value chain, the following methods were applied.

3.3.2.1 Materialization assessment

Materiality analysis is a tool used for identifying priority of sustainable issues considering the perspective of key stakeholders of a project [136, 137]. The method is characterized by its ability to differentiate between activities impacts and effort in perspective to the company. However, as that is the core of the methods, it often fails in considering deeper sustainability issues such as the environmental and social perspectives [136, 138]. Materiality assessment is most often used by companies to showcase financial stability for investors[139].

The method is highly criticized by its ability to not disclose operations done by the company, as while the method is successful in identifying, differentiating and classifying sustainable topics, it doesn't effectively evaluate

company execution on these topics [138].

The method begins with the identification of stakeholders, which may range from decision-makers within the company, responsible for strategic decisions, to external parties involved in the company's activities [136, 138]. For the purposes of this project, the stakeholders considered are detailed in Appendix D.

For this analysis, a single representative per department was invited to participate in the discussion. This approach was adopted to ensure an equal level of representation across departments. Allowing open participation could have distorted the results, as teams with greater engagement in the project would likely have higher attendance, thereby influencing the final prioritization and biasing outcomes toward their objectives. Additionally, the participation of external stakeholders in this discussion was limited due to confidentiality constraints. The following list states the identified key stakeholders for the mitigation actions materiality assessment:

1. Material engineers
2. Glass design engineers
3. Circular strategist
4. Glass buyer
5. Legal analyst

Methods used to evaluate the importance of identified sustainability actions/issues are typically based on surveys with scoring systems or through direct discussions with stakeholders [136, 138]. For the purposes of this project, the discussion session was conducted in person, using a Miro board for data collection and prioritization. Following this, a materiality matrix was developed to support the prioritization process. The X–Y axes represent different stakeholder categories, and the various zones are defined based on the level of importance assigned to each sustainability issue by the stakeholders. An example of this approach is illustrated in Figure 3.6.

During the session, each issue and its corresponding list of proposed mitigation actions were presented one by one. After each presentation, participants were given a few minutes to add new mitigation actions, if they

identified any additional ideas, and to begin prioritizing the actions on their boards. Once all participants had completed their work in Miro, the session was opened for discussion, allowing attendees to share the reasoning behind their decisions. These insights were documented and recorded to support the roadmap development process. Finally, a summary of the key discussion highlights was presented before moving on to the next issue. In total, five high-priority issues were discussed during the session.

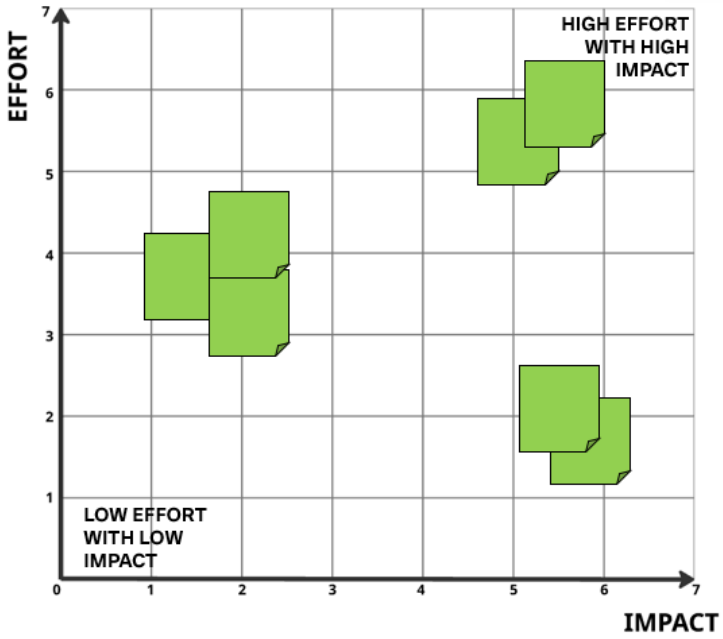


Figure 3.6: *Example of materiality matrix*

3.4 Reliability and Validity Considerations

3.4.1 Validity & Reliability of the method

Through this research, the aim is to deliver a relevant and valuable study for future readers. Therefore, a commitment is made to ensure that only accurate data, with clear traceability to its original sources, is presented throughout the research. To uphold appropriate research ethics, all stages of the study are thoroughly documented to enhance the transparency and replicability of the project. Furthermore, a high level of validity is pursued by clearly

referencing the origin, context, and basis of all information and findings, thereby strengthening the overall reliability of the research.

3.4.2 Validity & Reliability of Data

The insights collected from the literature were processed through an evaluation of the methodological rigor of the reviewed articles, with particular attention to the methodology sections in order to assess their reliability. In the case of this research, the data collected from interviews were interpreted following the approach proposed by Kvale (1996) [140], which suggests that interview statements should be evaluated based on their plausibility and coherence with empirical evidence. Accordingly, for a response to be considered reliable, it must be supported by credible sources and demonstrate consistency with the overall findings [140]. The reliability of data obtained from organizational websites was assessed by examining the potential benefits for the organizations and by considering whether there may be underlying motivations that could influence or bias the presented information.

Chapter 4

Results

In this chapter, the results of the methods applied are going to be presented.

4.1 ESG issues identified in the Automotive Glass Value Chain

Figure 4.1 illustrates the identified value chain of the automotive glass industry. Through the analysis of this value chain, two key aspects were identified.

The first aspect is the vertical integration of major suppliers across the first three stages, as these suppliers often own mining operations, float glass production facilities, and component manufacturing plants (as indicated by the red square highlighting the first three stages in Figure 4.1).

The second characteristic identified in the value chain is the low level of circularity within the automotive glass industry. This is illustrated by the presence of a circular loop between the float glass production and component manufacturing stages. This loop represents existing recycling practices implemented by suppliers, whereby internal production waste is reprocessed and reintegrated into their own manufacturing processes.

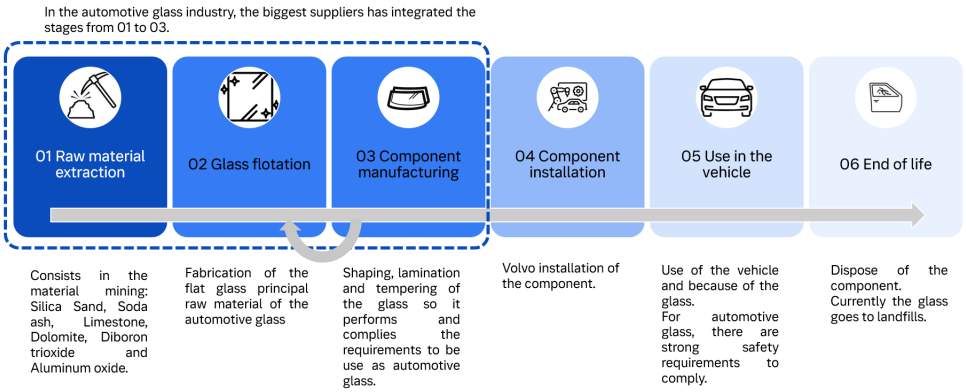


Figure 4.1: *Structure of the Automotive Glass Value Chain*

Following the findings from the literature review on ESG risks, additional ESG-related issues were identified and incorporated into the initial list. These new issues emerged from the stakeholder interviews conducted throughout the study. The identified issues are as follows:

1. **EU taxonomy on diboron trioxide:** During the interviews, respondent *S3* highlighted that diboron trioxide, a substance present in borosilicate glass used in applications such as camera lenses and sensors, is classified as a substance of very high concern under European Union regulations. Consequently, this material falls under the scope of the EU taxonomy, implying that products containing it may not qualify as sustainable. This observation was further supported by external sources, such as the European Chemicals Agency, which identifies diboron trioxide as potentially harmful, with evidence indicating adverse effects on fertility and possible risks to the unborn child during pregnancy. Based on these considerations, this issue was classified under both Social & Governance risks and Supply Chain risks.

2. **Oligopolistic market structure:** According to respondents *S1* and *S6*, the automotive glass market is characterized by an oligopolistic structure, dominated by a small number of large, vertically integrated suppliers. This level of integration across the value chain provides these firms with significant bargaining power over OEMs within the automotive industry. Market data supports this observation. In 2023, Fuyao Glass was identified as the market leader, holding approximately 22% of the global market share and supplying around 65 million square

meters of automotive glass. It is followed by AGC, with approximately 19% of the market share, while the third position is contested by Saint-Gobain Sekurit (automotive glass division) and NSG. This concentrated market structure presents several challenges, particularly in relation to competition and innovation. Historically, major suppliers have been involved in anti-competitive practices, including price-fixing agreements. A notable case involves Saint-Gobain, which was found guilty by the European Court of engaging in price fixing, customer allocation, and the exchange of confidential information within the automotive glass market. As a result, the company was fined 896 million euros in 2008, a penalty later reduced to 715 million euros in 2014. These observations reinforce the perception of the automotive glass sector as "competitive, difficult, and closed," as described by respondents *S1* and *S6*, and highlight its classification as a significant supply chain risk within the ESG framework.

3. **Trade tensions:** Another important consideration, highlighted by respondents *S6* and *S4*, relates to the current global geopolitical context. This factor, combined with European Union restrictions on information sharing as part of broader security strategies, has increasingly affected the automotive glass sector. Respondent *S6* noted that both automotive glass suppliers and OEMs are impacted, as importing goods from non-EU countries has become more complex and costly. As a result, sourcing decisions now require greater strategic consideration, particularly in relation to supply chain resilience and cost management.

4. **EU provisional agreements:** Another relevant finding identified during the development of the project relates to the proposed ELV regulations presented in February 2026. These proposals include two provisional agreements specifically concerning automotive glass. The first establishes a requirement for the mandatory dismantling of at least 70% of automotive glass components. The second aims to ensure that the dismantled glass is recycled back into new glass products by 2026. These proposed regulations highlight the increasing regulatory pressure toward enhancing circularity within the automotive glass industry and reinforce the need for improved end-of-life management practices.

Based on the automotive glass value chain, and the risks identified through both the literature review and stakeholder interviews, the resulting set of sustainability-related issues is presented in Figure 4.2. The figure illustrates all

identified issues across the value chain and classifies them according to a color scheme aligned with the three main ESSENZ+ pillars: Social & Governance, Environmental, and Supply Risks:

1. Environmental: (represented by green in Figure 4.2).
    (a) Silica sand mafias
    (b) Synthetic soda ash environmental impacts
    (c) Silica sand mining impact to fauna and flora
    (d) Energy intensive process, high CO_2 emissions, from use of fossil fuels
    (e) Risk of PFAS use in hydrophobic glass. Coating application
    (f) Waste from the automotive aftermarket
    (g) Low levels of circularity
2. Supply risk: (represented by red in Figure 4.2)
    (a) Silica sand mafias
    (b) EU Taxonomy on Diboron trioxide
    (c) Oligopoly market: Cases of Price fixing in 2007.
    (d) China trade tension
    (e) EU Provisonal agreements
    (f) Risk of PFAS use in hydrophobic glass. Coating application
3. Social & Governance: (represented by blue in Figure 4.2)
    (a) Silicosis disease
    (b) Silica sand mafias
    (c) EU Taxonomy on Diboron trioxide

As shown, several issues were classified under two or more pillars due to their multidimensional impacts. This classification reflects the interconnected nature of sustainability risks within the automotive glass value chain. For instance, silica sand mafias represent illegal operations that extract materials from unauthorized locations without governmental oversight or transparency.

As a result, they have significant impacts across all ESG dimensions. From a social perspective, they are associated with poor working conditions

and potential child labor practices. Environmentally, they contribute to land degradation and non-compliance with environmental standards. Additionally, they pose a supply risk, as companies sourcing from such operations face disruptions, legal risks, and reputational damage. Following this logic, similar classifications were applied to other issues such as the EU taxonomy on diboron trioxide and the use of PFAS in coatings. Both materials are subject to strict regulatory frameworks, thereby representing supply risks. In addition, diboron trioxide presents potential social and health-related concerns, while PFAS usage is primarily associated with environmental impacts. This multi-pillar classification approach enables a more comprehensive understanding of the systemic nature of sustainability risks within the industry.

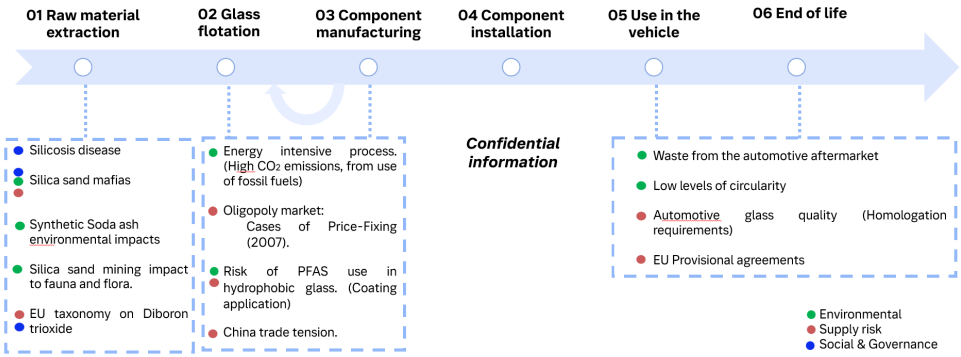


Figure 4.2: *Classification of the list of identified ESG challenges in the automotive glass value chain.Due to confidentiality stage 04, component installation was taken out, and due to the scope and similarity on the challenges stages 02 and 03, and stages 05 and 06 were joined.*

4.2 Technology and Innovation Opportunities

Based on the interviews conducted and the encoding method applied to the responses, four main findings were identified related to innovation practices and dynamics within the automotive glass industry (For tracking of the respondents check Table 3.1):

1. **SOI is dependent on business conditions:**

Sustainability initiatives are typically pursued when they align with economic viability and existing performance requirements, this statement was repeatedly detailed by the respondents *S1*, *S4*, *S5* and *S7*. Cost remains the dominant decision-making factor within the industry and plays a central role in most strategic and operational choices. Business conditions strongly shape innovation processes in the automotive glass market, a pattern that is reflected in where innovation activities primarily occur. In many cases, *S1* sustains that technological innovations are developed by the component manufacturer based on own initiative, who then incorporate these solutions into their portfolios and offer them to various clients. However, the adoption of such innovations by vehicle assemblers is not guaranteed, as it depends on factors such as cost implications and the ability to integrate these solutions into existing vehicle platforms. This dynamic illustrates a pattern in which assemblers tend to adapt to technologies proposed by suppliers, rather than acting as the primary drivers of innovation. Additionally, for the specific case of innovation for circularity, most of the interviewees suggested that this lack of innovation is due to the low value perceived in glass after its use phase. *S4* and *S9* mentioned that it is ”cheaper” to continue extracting raw materials, as there is currently ”no significant economic benefit in closing the loop”.

2. **Sustainable glass is a broad and complex concept:**

   Sustainable glass is commonly understood as low-carbon, circular, and recyclable; however, interpretations vary depending on departmental roles and levels of exposure to sustainability concepts, as indicated by respondents *S1*, *S2*, and *S3*. This variation results in uneven implementation across the organization. Departments with greater exposure to sustainability practices, such as those represented by respondents *S4*, *S5*, *S7*, and *S8*, tend to define sustainable glass not only in terms of reduced environmental impact, but also by incorporating social and governance considerations. More experienced respondents, such as *S6*, further extend this definition to include geopolitical and broader business-related sustainability dimensions. Overall, no clear consensus exists on the definition of “sustainable glass.” This lack of alignment can be attributed partly to differences in familiarity with sustainability terminology, as well as varying levels of exposure to the automotive glass industry itself. Many interviewees, particularly

those with advanced sustainability knowledge, noted that glass is not a material frequently addressed in their day-to-day work. This gap in exposure can lead to assumptions or an incomplete understanding of the actual challenges and realities associated with automotive glass.

3. **Collaboration is essential but limited:**

   Internal and external collaboration are recognized as critical enablers of sustainability-oriented innovation; however, they are constrained by factors such as bargaining power imbalances, limited time and resources, misaligned incentives, and coordination challenges across the value chain, this was expressed by *S1*, *S6* and *S9*. As identified in the value chain assessment, the automotive glass industry is an oligopolistic market, strongly dominated by suppliers, particularly in glass production and component manufacturing stages. Under these conditions, collaborative initiatives are highly dependent on supplier decisions, which can limit the ability of OEMs to effectively drive sustainability-related projects.

4. **The industry prioritizes exploitation over exploration:**

   Innovation efforts are primarily focused on incremental improvements, while exploratory and radical innovation activities occur selectively and typically require strong justification, additional resources, and clearly demonstrated business value. Although sustainability is strategically acknowledged, organizational ambidexterity is practiced in a conservative manner, with a strong bias toward short-term exploitation.

4.3 Identified High-Priority Issues and Proposed Mitigation Strategies

As detailed in the methods chapter, an S-FMEA was conducted to evaluate the criticality of identified risks based on their severity and likelihood of occurrence to then define corresponding mitigation actions. Through this analysis, from both the OEM and automotive glass industry perspectives, several risks were identified as having high priority scores, as presented in

Appendix E. The following provides a brief overview of the highest-ranked risks:

1. **Silicosis disease (300 Criticality score)**

   Silicosis is ranked highly due to its severity as a fatal occupational disease, exposing suppliers and OEMs to significant legal, human rights, and reputational risks. Although no specific cases have been identified among automotive glass suppliers within the scope of EU due diligence and ESG expectations, the responsibility of OEMs extends both upstream and downstream in the value chain. The risk is therefore considered high, despite the relatively low likelihood of occurrence due to existing safety measures. However, the enforcement of these safety standards varies across silica-producing regions, particularly in parts of Asia, Africa, and Latin America. Consequently, incidents continue to occur, especially in low-cost extraction environments.

2. **Silica sand mafias (300 Criticality score)**

   Although the risk is identified as geographically specific (e.g., India), and illegal extraction is typically associated with local use cases such as construction rather than high-quality applications like automotive glass production, the risk remains highly rated. This is because illegal mining practices undermine traceability, ESG compliance, and long-term supply security, thereby creating indirect but critical material risks for OEMs.

3. **Synthetic soda ash production environmental impact (100 Criticality score)**

   The high criticality score is primarily attributed to the elevated CO_2 emissions associated with soda ash production, which can reach approximately 1.2 tCO_2 per ton of produced soda ash. These emissions contribute significantly to environmental impact and directly affect OEM Scope 3 targets, as well as future carbon-related costs. Since the majority of soda ash production remains reliant on fossil fuels, this issue is widespread across the industry. It is important to note that synthetic soda ash has a higher environmental impact compared to natural soda ash, and is therefore considered a higher-risk material from a sustainability perspective.

4. **Silica sand mining impact to fauna and flora (100 Criticality score)**

A criticality score of 100 is assigned due to the significant environmental impacts associated with silica sand mining operations. Biodiversity loss resulting from these activities can lead to community opposition, delays in permitting processes, and increased risks of non-compliance with ESG requirements. Environmental disruption is inherently linked to mining activities unless extensive and effective mitigation measures are implemented.

5. **Extensive energy intensity process (100 Criticality score)**

   The risk is considered high because energy consumption drives cost volatility, contributes to significant CO_2 emissions, and increases exposure to carbon taxation. Failure to decarbonize production processes may therefore threaten the long-term competitiveness of the industry. Float glass furnaces are structurally dependent on high-temperature processes that currently rely heavily on fossil fuels. Although the glass sector is actively pursuing decarbonization strategies, with carbon neutrality targets set between 2040 and 2050, the industry remains largely dependent on natural gas. This dependence has exposed the European market in particular to energy supply risks and price volatility associated with regional geopolitical tensions.

6. **Oligopoly market (100 Criticality score)**

   The risk is considered high due to the concentrated structure of the automotive glass industry. Supplier concentration increases pricing power, reduces supply chain resilience, and limits the strategic flexibility of OEMs. The European market, in particular, is characterized by high entry barriers and significant capital intensity, further reinforcing supplier dominance and associated risks.

7. **Low level of circularity (100 Criticality score)**

   The issue is considered critical due to the limited availability of recycling infrastructure for laminated automotive glass. Current supplier key performance indicators (KPIs) indicate that only 1-8% of end-of-life glass cullet is recycled back into production. This raises significant long-term sustainability concerns, as it contradicts OEM sustainability commitments and exposes companies to potential regulatory constraints and reputational risks.

4.3.1 Proposed Mitigation Strategies

Through brainstorming and market research, a list of potential mitigation actions addressing the high-impact risks identified in the automotive glass industry was proposed for further discussion and to be subjected to the materiality assessment:

1. **Silica Sand mining risks**

   Under the risks associated with silica sand mining, including silicosis, illegal sand extraction, and impacts on fauna and flora, these issues were grouped together, and a set of combined mitigation actions was proposed.

   First, the implementation of certified sustainable mining requirements, combined with full raw material traceability for silica sand across the industry, is proposed. This includes the use of third-party certification schemes in collaboration with suppliers to ensure ethical and responsible sourcing practices.

   The second proposed action serves as an alternative to the first. It aims to verify the traceability of silica sand through the use of third-party investigators to audit and validate suppliers within the glass industry, ensuring that mining activities adhere to ethical and responsible practices.

   The third proposed action involves the implementation of a blockchain based system for tracing silica sand and ensuring responsible sourcing of the material. This approach would further strengthen Volvo Cars' commitment to ethical and responsible business practices, while enhancing transparency across the value chain. Volvo Cars already applies enhanced due diligence measures for Lithium, Nickel and Cobalt through its Responsible Sourcing Program.

   The fourth proposed action involves collaboration with silica sand suppliers to implement biodiversity protection and restoration practices. Suppliers could invest in techniques aimed at reducing environmental degradation and supporting land rehabilitation, thereby generating positive environmental outcomes. For instance, an EU-funded project

successfully restored 13 hectares of a silica sand mine in Spain, demonstrating the feasibility of such approaches [141].

The fifth proposed action is the recognition of silica sand as a raw material of concern (RMoC) within the industry. Such classification would increase awareness of the associated risks and elevate the priority of silica sand in sustainability considerations. The inclusion of silica sand in RMoC lists is not unprecedented; for instance, the Mercedes-Benz Group already classifies silica sand as a raw material of concern in its sustainability assessments [142].

2. **Soda Ash mining / production**

   The following mitigation actions are for the negative environmental, health and wellness impacts of synthetic soda ash combined:

   The first action involves the use of third-party investigators, to validate soda ash suppliers. This approach is not unfamiliar in the synthetic soda ash sector as in the past Solvay (Soda-Ash supplying company in US) did similar action in compliance with Canada's Fighting Against Forced Labor and Child Labor in Supply Chains. Through the investigation it was confirmed that no forced or child labor was happening through their supply chain.

   The second action is the request from automotive glass suppliers to use natural soda ash instead of the synthetic to reduce the environmental impacts. Synthetic soda ash is mentioned to produce 3 times higher carbon footprint than natural soda ash due to its energy intensity and pollution consequences. The use of natural soda ash instead of synthetic in the glass industry is not uncommon and efficiently applicable [143].

   The third action proposed is the implementation of third-party certification across the soda ash suppliers to ensure sustainable activities of both the suppliers of the mined soda ash but also the chemically developed synthetic soda ash.

3. **Extensive energy intensity process**

The first mitigation action implies for the OEMs industry to collaborate collectively for a funding of greener agreements in the automotive glass industry. Suppliers were already identified to have net-zero greenhouse gas emissions by 2040-50 however funding could accelerate implementation of the suppliers' strategies.

The second proposed action focuses on minimizing the use of glass and investing in alternative materials in order to reduce dependency and the associated energy-intensive production processes. This proposal encourages OEMs to identify and investigate emerging alternatives to conventional automotive glass. One example is polycarbonate glazing, which can offer a reduced carbon footprint, improved recyclability, and additional benefits such as lower weight compared to conventional glass. However, this material also presents limitations, including sensitivity to scratching and degradation under ultraviolet (UV) exposure, which can negatively affect its optical performance.

The third proposed action involves the adoption of carbon neutrality strategies to reduce the carbon footprint of automotive glass through carbon offsetting mechanisms. This approach also emphasizes the integration of automotive glass solutions that contribute to indirect emission reductions at the vehicle level, for example by improving energy efficiency in other components or systems, thereby lowering overall energy consumption. An example of such an approach is the integration of windshields with enhanced thermal insulation properties, which limit heat ingress into the vehicle cabin. This reduces the reliance on air conditioning systems, thereby decreasing energy consumption during vehicle operation and resulting in lower CO_2 emissions during the use phase.

4. **Oligopoly Market**

   For the oligopolistic market risk, a single mitigation action is proposed, whereby OEMs invest in emerging automotive glass suppliers in order to rebalance power within the supply chain. Rather than attempting to immediately replace incumbent suppliers, this approach emphasizes the development of long-term strategies, such as off-take agreements or co-development partnerships, to support smaller or technology-focused firms in scaling their capabilities and becoming credible suppliers for OEMs.

5. **Low levels of circularity**

   The first proposed action is to initiate innovation orchestration across the automotive glass value chain through the establishment of structured circular partnerships. This recommendation encourages OEMs, glass manufacturers, raw material suppliers, start-ups, universities, non-governmental organizations (NGOs), and recycling companies to collaborate across sectors and jointly develop sustainable solutions.

   The second proposed action involves developing a business case for closing the loop in automotive glass. The objective is to ensure that recovered glass is reintegrated into new float glass production, rather than being downcycled into lower-value applications such as glass fibers.

   The third proposed action involves initiating research into design for recyclability in automotive glass components, with the aim of enhancing material retention strategies.

4.3.2 Materiality matrix

Insight gathered from the workshop are as shown briefly in Figure 4.3 and more in detail in Appendix F. The results of the materiality matrix indicate that the initiative with the highest impact and effort is the diversification of suppliers. This action is considered beneficial; however, discussions held during the focus group revealed that this initiative is already underway, with continuous supplier assessments currently being conducted.

Other mitigation actions discussed elicited mixed responses, particularly those related to third-party certification schemes and the implementation of blockchain for silica sand traceability. While these measures were recognized as having high potential impact in improving traceability and ensuring sustainable sourcing practices, their implementation was perceived as highly demanding in terms of effort. This is primarily due to the reliance on supplier compliance, in an industry where suppliers possess considerable bargaining power due to limited alternatives.

Closing the loop with suppliers, as well as developing innovation orchestration initiatives involving aftermarket and recycling partners, were

identified as options with high impact and moderate levels of effort, mainly associated with negotiation processes and the development of robust business cases.

Finally, design for recyclability and the recognition of silica sand as a RMoC were both identified as high-impact, low-effort initiatives, with the potential to generate significant benefits for the sustainability of the automotive glass industry.

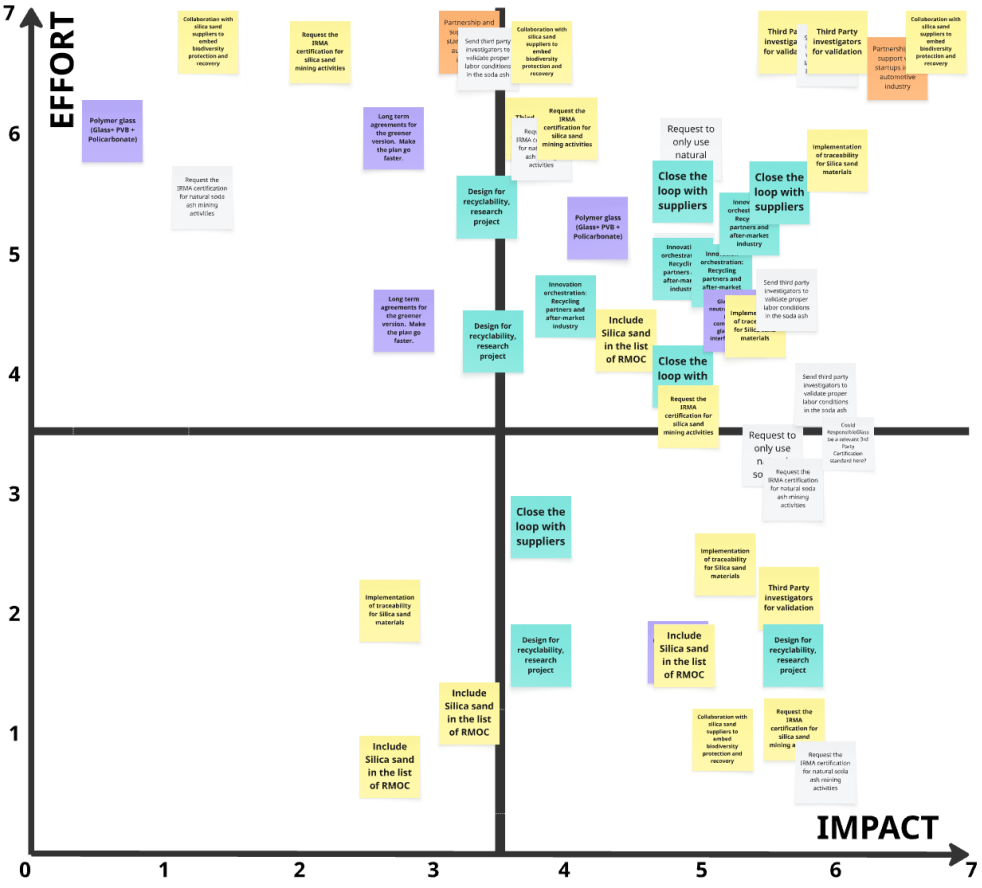


Figure 4.3: *Materiality Matrix Results*

Chapter 5

Discussion

In this chapter, the discussion of the project will be presented, considering the identified ESG risks, their relationship with SOI practices and dynamics, as well as with sustainable sourcing. Finally, the recommendations and mitigation actions introduced in the previous chapter will be discussed in greater detail.

5.1 Sustainability Challenges Across the Value Chain

An analysis of the automotive glass supply chain reveals a clear polarization of sustainability impacts and operational challenges when comparing upstream and downstream operations.

5.1.1 Upstream Dynamics: Raw Material Extraction and High-Impact Manufacturing

The upstream segment can be divided into two primary phases: raw material extraction and the industrialized Tier 2 and Tier 1 manufacturing phases. These manufacturing phases include float glass production, shaping, tempering, and lamination of the component.

The raw material extraction phase emerges as the most risk-intensive stage across the entire value chain, accumulating the highest concentration of social, environmental, and supply-related risks. This concentration of impacts

aligns with established supply chain literature, which notes that Tier 3 and Tier 4 extraction processes often carry the most significant environmental and socio-economic burdens [144, 69]. To address these systemic vulnerabilities, particularly those associated with silica sand mining and soda ash production, regulatory frameworks and sustainable mining initiatives are increasingly being promoted by governments. An example of this is the European Union's regulatory approach, including directives and frameworks such as the EU taxonomy [145, 146].

The subsequent upstream phase, focused on glass manufacturing and processing, presents a risk profile primarily centered on environmental and supply chain constraints. Environmental concerns are mainly associated with intensive energy consumption, particularly in furnace operations, as well as the potential use of chemical substances, such as PFAS, which are increasingly subject to regulatory restrictions due to their adverse impacts on human health and ecosystems. Similar concerns apply to materials such as diboron trioxide, which is regulated under the EU taxonomy framework.

From a supply chain perspective, these environmental risks are closely linked to geopolitical conditions. In the case of glass production, the reliance on fossil fuels, predominantly natural gas according to respondents *S6* and *S9*, to power manufacturing facilities exposes the value chain to geopolitical volatility. This directly threatens revenue performance and organizational resilience. This point was reinforced by respondent *S6*, who described how geopolitical instability affects quotation processes and creates tensions with the company.

This connection highlights a critical insight: sustainability should not be understood solely as an environmental objective, but also as a business requirement. In this specific case, transitioning toward alternative and cleaner energy sources can serve as a strategic mechanism to reduce dependency risks and strengthen organizational resilience under conditions of uncertainty [147, 148].

5.1.2 Downstream Dynamics, Regulatory Pressures and the Circularity Gap

From another perspective, downstream impacts are primarily characterized by environmental compliance and supply security risks. This downstream risk profile is shaped by two main factors.

First, there is a strong governmental push toward increasing circularity through emerging legislation. The current approach of treating the automotive glass value chain as a linear system is becoming obsolete and increasingly misaligned with modern industrial strategies [149, 150]. Second, these legal frameworks are a direct response to the environmental impacts caused by end-of-life component disposal and the accumulation of post-consumer waste. This represents a double inefficiency: valuable materials with the technical potential to be reintroduced into a closed-loop system as glass can theoretically be fully recycled continue to be landfilled or downcycled into lower-value applications. As a result, the industry remains dependent on the continuous extraction of virgin materials, thereby increasing upstream environmental impacts.

Currently, the automotive glass value chain remains linear, showing the low levels of circularity. While this environmental risk has not been changing by the market in recent years, there is a new risk of non-compliance due to upcoming European regulations, that could presents a profound opportunity. Provisional agreements and new circular requirements should not be viewed only as operational threats or survival effort; rather, they must be interpreted as critical external drivers capable of enhancing the industry's level of sustainability.

The fact that these regulatory proposals are new, with less than 6 months (February 2026 [151]), reinforces that circularity within the automotive glass sector is still in its early stages. However, it also indicates that the focus has expanded beyond the attention of individual OEMs (like Volvo Cars) this issue has brought the attention to macro level regulatory bodies. These institutional actors possess the necessary leverage to shift the supplier bargaining power and mobilize this oligopolistic market towards to a systemic sustainable innovation.

5.2 State of Sustainable-Oriented Innovation

SOI was identified as being present within the automotive glass sector. However, it is primarily driven by OEM suppliers, particularly in relation to component and process improvements. The development of SOI is largely constrained by prevailing business conditions, as the sector is characterized by low circularity, which is partly attributed to limited economic incentives.

The finding that SOI is dependent on business conditions is considered highly reasonable. While the development of genuinely embedded SOI, incorporating environmental and social dimensions, is desirable, it is essential to account for the economic viability of such solutions. Economic performance represents a fundamental pillar of sustainability, alongside environmental and social aspects [86]. This observation is applicable across SOI practices in various materials and industries, and is not limited solely to the automotive glass sector.

SOI theory emphasizes the importance of adopting a systems-level thinking perspective, within which the concept of circular economy is highly relevant. Circular economy offers multiple benefits at both system and value chain levels by narrowing, slowing, and closing resource flows, thereby retaining value from products, materials, and activities over time [101].

Incorporating circular economy within the automotive glass industry was identified to be particularly challenging due to the complexity of profitability in the system perspective. Furthermore, the concept of sustainable glass was found to vary across actors within the value chain, indicating a lack of shared understanding or alignment.

From a broader perspective, and when benchmarking against other materials, there is a higher level of innovation maturity in materials such as aluminum and batteries. There are already examples of implementing innovation orchestration and circularity in this cases, as detailed by *S4* and *S5*. However, compared to these industries, the glass industry is not at the same level of maturity. This situation also reflects that inside the organization, Volvo Cars, the maturity level of SOI differs from material to material, depending on the specific sector of the business the team is focused.

Considering these findings, and through a comparison with Adams et al. (2016) framework for assessing the maturity level of SOI, the automotive glass industry can be characterized as operating at a reactive, incremental, and internally focused level. While this represents a partial adaptation of the original model which was intended to be applied at the firm level rather than at the industry level it still provides a useful lens for interpretation. The automotive glass industry is perceived as primarily adopting an operational optimization, or "eco-efficiency," approach. Current innovation practices are largely centered on "doing the same things better" resulting in incremental reductions in environmental impact rather than transformative changes [86].

Innovation efforts are predominantly focused on incremental improvements, while exploratory and radical innovation activities occur selectively and typically require a clearly demonstrated business value to be pursued. Although sustainability is strategically acknowledged, organizational ambidexterity is practiced in a conservative manner, with a strong bias toward short-term exploitation. This tendency constrains the automotive glass industry's ability to achieve more transformative and long-term sustainability outcomes.

While many reasons to why this is happening in the automotive industry were identified such as quality requirements, profitability and legislation were identified. SOI theory recommends to take a more transformative approach to achieve long term competitive advantages [86, 24].

Key drivers for SOI at a transformative level include the need for increased cross-sector collaboration, enabling innovation from a systems-level perspective [23]. Although both internal and external collaborations are recognized as critical enablers of SOI, they are significantly constrained by factors such as supplier bargaining power. As identified in the value chain analysis, the automotive glass industry is characterized as an oligopolistic market, strongly dominated by suppliers in the glass production and component manufacturing stages. This structure creates a high level of dependency on suppliers for decision-making related to sustainability initiatives.

However, from an OEM perspective, Volvo Cars has the opportunity to place greater emphasis on the design phase, where it has direct influence

and the potential to significantly impact material circularity and overall sustainability performance [103].

5.3 Implications for Sustainable Sourcing and Supply Chain Management

When evaluating Volvo Cars' operations through the lens of Green Supply Chain Management (GSCM), the organization demonstrates several characteristics of sustainable value chain management [152, 62]. Data gathered from stakeholder mapping and interviews revealed a structured internal governance framework. This was evident through the presence of a dedicated Material Strategy Team, which functions as a link between design-related engineering departments, such as materials engineering and design, and sustainability units. This cross-functional integration enables the organization to evaluate and incorporate sustainable alternatives into its material portfolio, while balancing environmental and social impacts with organizational value creation [69].

Additionally, a specialized Sustainable Procurement Team operates from a commercial and business development perspective. This team actively develops purchasing strategies that aim to generate simultaneous benefits for the environment, society, and the corporation's long-term competitive advantage [153]. The institutionalization of these specialized teams and organizational practices provides clear evidence of an established responsible sourcing culture within Volvo Cars' organizational structure. However, given that the firm manufactures highly complex products, namely high-technology vehicles supported by multi-tiered global supply chains, it is not possible to classify the entire corporation under a single responsible sourcing archetype.

For the specific case of the automotive glass sector, the empirical data suggest that this sub-chain currently operates under the Social Risk Manager archetype [154, 69]. Within these boundaries, Volvo Cars primarily focuses on mitigation and compliance-driven monitoring to manage immediate ESG risks, rather than actively transforming the market or standardizing proactive innovations.

This posture is largely explained by the fact that circular glass regulations

remain in an early stage of legislative development. Consequently, the organization are adopting a protective approach aimed at ensuring supply continuity and avoiding risk, without committing significant innovation capital to radical transformation within this specific material category.

On the other side, when evaluating Volvo Cars from a broader organizational perspective, particularly regarding materials such as aluminum, copper and cobalt. Is possible to see that the organization transitions into a mixed archetype combining characteristics of a Compliance Network, a Collaborator, and a Proactivator [155]:

1. **Compliance Network:** Volvo Cars applies mature strategies to comply with existing international regulations for these materials, while actively participating alongside other OEMs in cross-sector initiatives aimed at establishing industry standards for responsible sourcing [156].

2. **Collaborator:** In certain material categories, Volvo Cars adopts a collaborative and adaptive approach by aligning its strategies with the sustainability maturity and proactive initiatives driven by dominant suppliers or peer OEMs [157].

3. **Proactivator:** There is both historical and contemporary evidence of Volvo Cars acting as an industry frontrunner. This is demonstrated by corporate decisions such as making the three-point seatbelt patent freely available to improve global traffic safety, as well as by being an early adopter of blockchain-based traceability solutions for critical raw materials [158].

This dynamic highlights that, while Volvo Cars actively pursues sustainability leadership and promotes responsible sourcing at the corporate level, the maturity of SOI varies across material streams. The automotive glass value chain represents a relatively overlooked area, more closely aligned with a Risk Manager approach than with the proactive and collaborative frameworks observed in the company's core strategic material categories.

5.4 Interconnection Between ESG Risks and Innovation Dynamics

As presented in the results chapter, a summary of sustainability risks was identified across the automotive glass value chain. Among these, the S-FMEA

highlighted eight critical risks, which can be categorized according to the ESSENZ+ framework as follows:

1. **Social & Governance:** Silicosis disease and silica sand mafias.
2. **Environmental:** Silica sand mafias (ecosystem disruption), the environmental footprint of synthetic soda ash production [147], the environmental impact of silica sand mining [146], and low levels of circularity [159].
3. **Supply Chain Risk:** Oligopolistic market structure [159, 160].

These three aspects, according to our analysis, have different work dynamics and management practices. Each of these groups has its own considerations, status, level of maturity, and degree of knowledge acquisition. Each of them is detailed and explained in the following subsections.

5.4.1 Social and Governance Risks, maturity and Reactive Innovation

Although social and governance risks received the highest criticality score (Criticality = 300) due to their direct and severe impacts on human health and safety, these areas also demonstrated the highest level of industry awareness, as well as the most developed mitigation practices and accumulated knowledge. Several standards governing mining and chemical production already exist to regulate these aspects, such as the Initiative for Responsible Mining Assurance (IRMA) and various sustainable chemical certification schemes, which are actively applied by Volvo Cars and other stakeholders within the automotive industry [161, 155].

This maturity in risk response is also reflected in the industry's approach to other critical raw materials, such as the implementation of blockchain-based traceability systems for cobalt and lithium [158]. Volvo Cars integrates such requirements directly into its supplier nomination process, as well as within its Green and Responsible Sourcing strategy through the Supplier Code of Conduct [162].

Additionally, because social risks are transversal across multiple raw materials, the organization's capability and innovation maturity in mitigating or preventing such risks is relatively high. This systemic awareness extends

beyond Volvo Cars, as competing OEMs, such as Mercedes-Benz and the Volkswagen Group, have adopted similar approaches through their own Materials of Concern lists and RMoC frameworks [163, 164]. This evidence suggests that the automotive industry is responding in a relatively consistent manner to address similar social and governance risks across raw material value chains.

5.4.2 Environmental Risks, the trade-off between Incremental Innovation and Circularity

On the other hand, environmental risks, despite receiving a lower criticality score of 100 compared to the score of 300 assigned to social risks, remain largely under-addressed, indicating a significant lack of circular innovation. While some suppliers demonstrate incremental improvements in energy efficiency, as illustrated by the examples shared by respondent *S9*, such as the gradual deployment of solar energy to reduce reliance on internal fossil fuel consumption, the transition toward true circularity remains limited.

Data from interviews with an automotive glass recycling organization revealed that, although recycling is technically feasible, the business model and operational dynamics have remained largely unchanged over the past six years. This indicates a very slow innovation cycle, primarily driven by the limited perceived economic value of returning post-consumer glass to suppliers. While the recycling supplier attributed this stagnation to the limited value associated with recycled glass cullet, academic and industrial literature primarily emphasizes the technical difficulties of the recycling process itself, particularly in relation to reintegration into float glass production [165, 166]. This lack of consensus in defining the problem may hinder systemic progress toward improved circularity.

From a methodological perspective, environmental risks may not have emerged as top priorities in the FMEA because the criticality matrix was calculated strictly based on Severity and Occurrence. The assessment did not include a Detection or Actionability parameter, nor any factor capable of capturing the current state of mitigation maturity or technological readiness [167, 132]. Reflecting on this limitation suggests that, although an environmental risk may appear less severe than a social risk within the scoring matrix, lower levels of awareness, knowledge, and mitigation maturity

can make such risks more challenging from an innovation and sustainability performance perspective.

5.4.3 The Oligopolistic Market, moderator and driver of Innovation

Finally, the oligopolistic nature of the automotive glass market represents a complex supply chain risk. Rather than constituting a fixed constraint, this market structure acts as a moderating factor that can either enable or restrict innovation, depending on the strategic posture of the organization [168, 169].

An oligopolistic market structure can strongly influence innovation when a limited number of global suppliers leverage their market concentration to resist customization or sustainability-oriented improvements. In such cases, OEMs may be forced to accept standardized, non-circular components due to the lack of viable alternatives. Conversely, this market structure can also act as a driver of innovation if managed through long-term strategic alliances. In highly concentrated markets, firms often mitigate risks by developing co-development agreements, thereby transforming supplier constraints into potential technological advantages [170]. However, it is important to recognize that these opportunities also involve specific risks and implementation requirements. Successful collaboration depends on factors such as appropriate governance structures, effective knowledge sharing, clear communication, and alignment of strategic objectives across the involved actors. As a moderating factor, oligopoly indirectly influences several innovation drivers, including:

1. **Cost:** Market concentration influences pricing mechanisms for sustainable alternatives, often increasing the initial investment required for low-carbon or more sustainable glass formulations [168].

2. **Perceived value:** Suppliers operating within an oligopolistic market tend to prioritize innovations that ensure high-volume financial returns. As a result, sustainability-oriented solutions often need to be linked to improved performance, regulatory compliance, or premium brand positioning in order to be perceived as valuable [171].

3. **Knowledge concentration:** Technical expertise related to glass chemistry, furnace electrification, and process decarbonization is concentrated among a limited number of global suppliers. Consequently, OEMs cannot innovate in isolation and remain dependent on suppliers' willingness to share knowledge and integrate new solutions [172].

4. **Governance agreements:** Overcoming bargaining power asymmetries in an oligopolistic market requires a shift from traditional transactional procurement toward robust collaborative governance frameworks. Such frameworks should incentivize mutual long-term sustainability objectives and support joint innovation efforts [169, 172].

5.5 Sustainable Road map development

Figure 5.1 showcase a proposed roadmap developed based on value chain, SOI analysis, and materiality assessment. Several insights were identified regarding potential recommendations and work streams. These insights supported the development of the following set of actions for the automotive glass industry. The proposed roadmap consists of a pre step and three main streams, each representing a key area of focus to improve sustainability performance within the automotive glass value chain:

1. **Governance & Vision:** Based on the interviews conducted, a key finding was the lack of alignment across departments regarding the definition and expectations of sustainable glass. This stream focuses on establishing clear organizational objectives and goals, with the aim of aligning internal perspectives and defining a coherent strategic direction for sustainability within the organization. This stream is identified as the initial phase of the roadmap. In contrast to the other streams, it is considered of primary importance, as developing a shared vision and a common definition of sustainability within the organization is a prerequisite for effective implementation. Additionally, it involves the formation of dedicated task forces to support and coordinate sustainability improvements across departments.

2. **Sourcing:** As identified in the value chain analysis, raw materials present significant social, environmental, and supply-related risks. To mitigate these impacts, this stream focuses on improving the

sustainability of sourcing practices for key materials such as silica sand and soda ash. This includes strengthening supplier engagement and capability development, as well as increasing transparency of material flows. The objective is to enhance collaboration with suppliers in order to improve sustainability maturity and promote responsible sourcing practices. Proposed short-term actions include the inclusion of these materials in RMoC lists, as well as a transition toward the use of natural soda ash, which generates significantly lower CO_2 emissions compared to synthetic alternatives. Long-term actions involve collaboration with certification schemes, such as IRMA, to ensure ethical business practices across the supply chain, along with continuous validation mechanisms within the industry. Furthermore, investments in biodiversity restoration initiatives are recommended as part of broader strategies for environmental mitigation.

3. **Technology:** This stream adopts a more exploratory approach, focusing on innovation in materials and component design. It includes the development and testing of new solutions based on concepts such as design for disassembly and recyclability. A proposed initiative within this stream is a dedicated recyclability project, in which KPIs are defined around increasing the quality and usability EOL glass cullet. The objective of this effort is to enhance material circularity and functional optimization, thereby improving material efficiency and enabling circular business models. In the longer term, continuous monitoring and investigation of emerging technologies are recommended to ensure that the organization remains aligned with technological advancements and innovation trends specifically in polycarbonate materials.

4. **Collaboration:** As highlighted in the analysis, the automotive glass industry is highly cost-driven and competitive, making collaboration essential for advancing sustainability and increasing the level of innovation maturity. This stream emphasizes the importance of cross-organizational partnerships, including the identification and mapping of potential collaborators, as well as participation in external initiatives aimed at improving sustainability performance across the industry. In the short term, this involves conducting research to identify potential partners within the automotive ecosystem that share similar values and objectives. In the longer term, it includes the development and implementation of collaborative business models and circular systems, with the aim of increasing the percentage of circularity within the

automotive glass value chain.

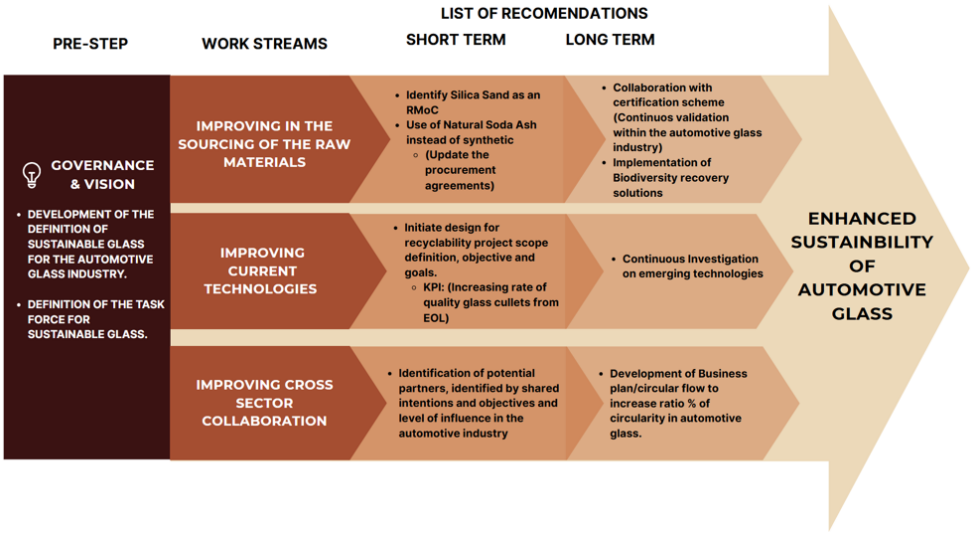


Figure 5.1: *Sustainable Road map proposed*

5.6 Feasibility of sustainability roadmap

As previously mentioned, the aim of the proposed sustainability roadmap is to enhance the sustainability performance of automotive glass. The roadmap targets key pain points identified throughout the study, including raw material-related risks, recyclability and circularity challenges, the lack of a shared understanding of sustainability, and low levels of collaboration across the value chain. While the roadmap aims to improve these areas, the extent of its impact remains uncertain.

Although the proposed actions are considered strategically relevant for generating both short-term and long-term sustainability benefits, their actual impact may be constrained by cost considerations and the oligopolistic structure of the automotive glass market. As discussed previously, decision-making power is largely concentrated among suppliers, which can make the implementation of sustainability improvements more challenging. However, an overview of major suppliers indicates that several key actors have established net-zero carbon emission targets for 2030, 2040, or 2050, as well as commitments related to ethical business practices. This suggests a willingness

within the industry to move toward improved sustainability performance.

The impact of these actions is expected to be significant across social, governance, and environmental dimensions. Improving circularity and establishing circular material flows could substantially reduce both CO_2 emissions and the extraction of virgin raw materials. For instance, one ton of recycled glass cullet can replace approximately 1.2 tons of raw materials and avoid up to 0.7 tons of CO_2 emissions [173]. This not only reduces environmental harm, but also mitigates social and governance risks by decreasing dependence on key virgin raw materials.

The assumptions regarding the roadmap time frames were based on the materiality assessment conducted with key Volvo Cars stakeholders. This raises the question of whether the roadmap is tailored from the perspective of Volvo Cars or from the broader perspective of the automotive glass industry. While the value chain analysis and the identified issues were developed from an automotive glass industry perspective, the assessment and prioritization of these issues were conducted from an OEM perspective.

This approach was primarily chosen due to the availability of stakeholders within Volvo Cars and because the study was conducted at the request of the organization. However, it would have been valuable to compare these findings with perspectives from external stakeholders. Extending the research externally could have enabled the collection of multiple stakeholder perspectives on the proposed solutions and mitigation actions.

Chapter 6

Conclusions and Future work

6.1 Conclusions

6.1.1 How is the current automotive glass value chain configured and to which sustainability risks is it exposed?

Through this study, the automotive glass value chain was mapped in considerable detail. In addition, CO_2 emissions and material flows were assessed for the key raw materials used in automotive glass. Several risks were identified throughout the study, particularly those related to responsible and ethical business practices, as well as circularity. These risks were further defined and ranked according to their criticality and sustainability impacts on the sector. The most critical risks for the automotive glass industry were identified within the social and governance dimensions. Regarding risk maturity, different levels of preparedness were observed across the identified issues. The findings suggest that environmental risks remain particularly challenging to address, especially under the current organizational mindset, which is primarily oriented toward incremental innovation.

6.1.2 How are sustainability-oriented innovation practices applied in practice compared to their theoretical framework?

Through this study, an in-depth assessment of SOI practices and maturity levels was conducted in order to compare theoretical perspectives with

industry practice. The results indicate that the automotive glass industry demonstrates an ambition to improve the sustainability performance of automotive glass. However, such improvements remain highly dependent on component manufacturers, who currently act as the primary decision-makers in relation to innovation and sustainability initiatives. To achieve long-term and meaningful sustainability improvements, stakeholders across the automotive glass sector need to strengthen collaborative innovation mechanisms. This is essential for shifting the industry from a predominantly absorptive role toward a more proactive and technology-driven position. Furthermore, stakeholders within the system should adopt a systems-level and circular-thinking approach, which extends beyond circularity alone to also consider the design, production, use, and end-of-life stages of automotive glass.

6.1.3 What actions can the automotive glass industry take to enhance its sustainability performance?

Throughout the study, a roadmap was developed and proposed from the perspective of the automotive glass industry, incorporating mitigation actions across the value chain to address the identified key challenges. The roadmap is designed to include both short-term and long-term actions, with the objective of shifting from a compliance-driven approach toward a more proactive stance in managing sustainability risks as identified thorough the SOI assessment. These actions are proposed with the intentions of developing stronger sustainable business practices, taking into account environmental, economic and social perspectives.

6.2 Limitations

While this research provides valuable strategic insights for Volvo Cars and the broader automotive glass industry, several inherent limitations must be acknowledged to contextualize the findings.

1. Stakeholder engagement and sample representation The primary empirical data of this study relied on qualitative data gathered from internal, external interviews and a dedicated focus group. For that reason, the depth of the analyzed risks and opportunities is tied to the subjective perspectives, level of engagement, and institutional

knowledge of the stakeholders. Although the research successfully incorporated crucial external perspectives, like third-party certification schemes, suppliers, and environmental specialists. Expanding the sample size; to include a bigger sample of tier-1 and tier-2 suppliers would have provided a better visualization of the upstream value chain. The concentration of the glass market (oligopoly) restricted access to certain supply chain actors, meaning some localized operational constraints might remain unrepresented.

2. Methodological scope and data sourcing: The investigation used two approaches, initially gathering baseline sustainability data from open source literature like corporate sustainability reports, and public Life Cycle Assessments (LCAs), which were later validated and improved by stakeholders' interviews. A limitation of this approach is the reliance on self-reported data from sustainability reports from suppliers, which may possess different degrees of transparency or methodological consistency. Additionally, due to the 30-credit scope of this Master's thesis, it was not feasible to perform an independent, quantitative validation of the solutions proposed (which could include benefits and reduction of % of the environmental impact or suppliers qualification rates) or a rigorous financial cost-benefit analysis of the proposed roadmap technologies.

3. Time Constraints and Horizon Boundary Time acted as a significant constraint throughout this research, necessitating a strict cutting point on when to conclude data collection and transition to analysis. Larger supply chain investigations and regulatory impact assessments, like as those conducted by industrial organizations for example the European Commission, typically requires several years of research. Conducting a similar assessment within the accelerated time space of an academic thesis forced a strategic trade-off, shifting the focus from an exhaustive, quantitative supply chain assessment to a systemic and qualitative innovation framework. Therefore, the developed roadmap represents a strategic baseline that requires continuous iteration as market and regulatory schemes evolve.

6.3 Future Work and What Has Been Left Undone

While this study establishes a great baseline for the automotive glass value chain, several strategic and empirical dimensions remain open for future investigation:

1. **Feasibility and Financial Assessments:** Future research should focus on conducting detailed financial and operational feasibility assessments for the actions proposed in the roadmap. This includes quantifying the precise capital definition required for a succesful implementation.

2. **Cross Benchmarking in Complex Product Systems:** It is highly recommended to contrast these results with recognized global leaders in green supplier management within complex or consumer product systems, such as Patagonia, Ikea, or H&M, examples mentioned in the Literature review. Although these organizations face less rare raw materials dependency in their respective industries in comparison to automotive industry, it is important to highlight that their governance models for supply chain transparency are highly mature. Additionally, more comparative studies against recent, and changing cases like Tesla's vertical integration of battery supply chains, would be interesting literature for comparison.

3. **Organizational SOI Assessment:** The SOI diagnostic conducted in this thesis was bounded to a single material stream (glass). Future efforts should deploy a transversal organization assessment to fully map Volvo Cars' SOI maturity, evaluating across all procurement and engineering divisions and also to identify and evaluate how to achieve the same maturity level in all the organization.

4. **Corporate Group Dynamics (The Geely Perspective):** This thesis analyzed corporate dynamics primarily from the perspective of Volvo Cars as an individual entity, considering its bargain power, dynamics and more. However, as part of the broader Geely Holding Group, it would be very valuable for future studies to investigate how collective group-level procurement strategies, shared supplier networks, and overarching governance policies influence or restrict localized sustainability initiatives.

5. **Prototyping and R&D Hosting:** An interesting next step for the organization would be to host low-tier, exploratory R&D projects focused on the redesign and deconstruction of automotive glass architecture. Questioning the current configuration of the component (for example design for disassembly, alternative laminates, or re thinking the functionability of the components) is essential to increase the radical disruptions and gradually build sustainable, internal technological knowledge.

6.4 Sustainable Development Reflections

The thesis contributes to the the following United Nations (UN) Sustainable Development Goals (SDGs):

1. **SDG 8 – Decent Work and Economic Growth:** This thesis contributes to promoting sustainable economic growth within the automotive glass sector by recommending the adoption of circular economy practices and highlighting of SOI oractices across the industry.

2. **SDG 12 – Responsible Consumption and Production:** The study aims to support sustainable consumption and ethical production practices throughout the automotive glass value chain. This is reflected in the proposed roadmap, which includes actions across sourcing, production, and aftermarket stages.

3. **SDG 13 – Climate Action:** The thesis highlights environmental risks associated with automotive glass, including the disclosure of CO_2 emissions across its value chain. The proposed recommendations aim to reduce emissions and support climate-related action within the industry.

4. **SDG 15 – Life on Land:** The study emphasizes the impacts of land degradation resulting from raw material extraction as well as waste disposal at end-of-life. The proposed roadmap aims to mitigate these impacts and contribute to the protection and restoration of biodiversity.

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Appendix A

Materials analysis

A.1 Float glass primary materials

As previously discussed, the primary raw material used in automotive glass is float glass. This type of glass is typically composed of approximately 70% silica sand, 15% soda ash, and 15% a combination of limestone and dolomite, which serve to stabilize the mixture of silica sand and soda ash. Another commonly used raw material is recycled glass cullet, which helps reduce the consumption of silica sand and soda ash. However, its usage is limited, as higher proportions can negatively affect the quality of the final glass product.

Silica Sand: Silica sand is a granular material primarily composed of silicon dioxide (SiO_2). To be classified as silica sand, the material must contain more than 95% silica, along with small amounts of aluminum oxide, limited iron oxide, and low levels of other impurities [174]. Silica sand is an abundant material that is widely available across multiple regions globally. Market analysis of global production indicates a distribution of approximately 40% in North America, 25% in Europe, 20% in the Asia-Pacific region, and 15% in the Middle East and Africa [175].

Soda Ash: Soda ash is the commercial name for sodium carbonate (Na_2CO_3) and is widely used in various industrial applications, including glass manufacturing as well as detergent and soap production [176]. This material can be obtained either through natural extraction or synthetic production, with significant implications for its sustainability performance. Natural soda ash is derived from trona mining, whereas synthetic soda ash is produced from salt and calcined limestone [176]. Soda ash is the second most

utilized material in glass manufacturing. Its natural form are less abundant than silica sand with trona reserves primarily concentrated in regions such as China, the United States, and Turkey [177].

Limestone: Limestone is a naturally occurring material composed primarily of calcium carbonate and is one of the most abundant industrial minerals on Earth. Many limestone deposits are relatively pure, containing less than 5% non-carbonate impurities. Limestone can occur in various forms and may be classified based on its origin, chemical composition, structure, and geological formation [178].

Dolomite: Dolomite is a naturally occurring mineral formed through the process of dolomitization, in which limestone is altered by magnesium-rich fluids over extended geological periods [179]. Dolomite constitutes approximately 2% of the Earth's crust and occurs alongside minerals such as calcite and aragonite [179].

Recycled Glass Cullet: Although glass can theoretically be recycled indefinitely without degradation in quality, this is rarely achieved in practice [180]. This is primarily due to high collection costs and contamination issues arising when glass is broken and mixed with other materials. In addition, the automotive industry imposes stringent quality requirements, necessitating recycled glass cullet of very high purity with minimal contamination. As a result, the majority of end-of-life float glass is either disposed of in landfills or significantly downcycled into lower-value applications, despite the existence of recycling technologies [180]. In contrast, the recycling efficiency of internal and pre-consumer cullet is relatively high, due to the higher purity of the material and the advanced internal recycling capabilities of glass manufacturers, which allow cullet to be reintroduced into the furnace [180].

A.2 Recycled Content

The use of recycled glass cullet offers significant environmental benefits, including reductions in CO_2 emissions and raw material consumption [173]. Specifically, one ton of cullet can replace approximately 1.2 tons of raw materials and avoid up to 0.7 tons of CO_2 emissions [173]. Recycled glass cullet is collected from waste glass and undergoes processes such as quality control, separation, crushing, and sorting according to quality [173]. There

are 3 types of recycled glass culets as shown in Figure A.1. Internal cullet is generated and collected directly at float glass production facilities. Pre-consumer cullet is obtained from customers, such as glass processors, where flat glass is transformed into components (e.g., laminated glass). Both internal and pre-consumer cullet originate from production waste due to cutting, cracking, or quality control rejections. End-of-life cullet, in contrast, is generated from the disposal of glazing products at the end of their service life [173].

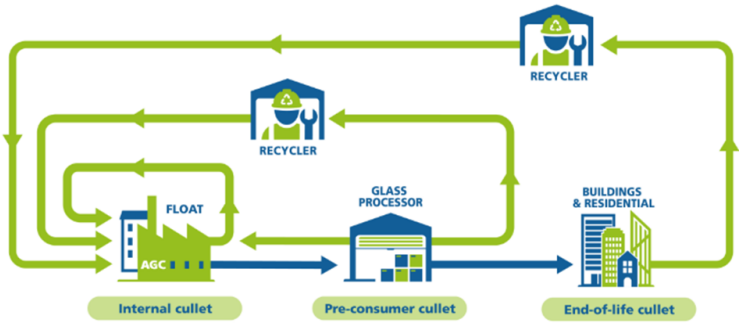


Figure A.1: *Float glass waste and recycling diagram [173]*

A.3 Glass Material Flow

A more detailed assessment of glass material flows is presented in a 2014 analysis, which illustrates the glass value chain from raw material extraction to end-of-life management, as shown in Figure A.2 [181]. China is identified as the largest global producer of glass, followed by Europe in second place [181].

The material flow diagram indicates that circularity is more commonly achieved in container glass, whereas flat glass demonstrates relatively limited circularity, primarily restricted to the reuse of cullet during the melting stage. The analysis further shows that, out of 92 Mt of silica sand, 20 Mt of limestone, and 32 Mt of soda ash, approximately 4 Mt end up in automotive glass components [181].

Recent developments in the automotive glass sector indicate gradual improvements in circularity. For example, AGC has established a partnership

with the recycling company Reiling to recycle approximately 300,000 pre-consumer windshields. This initiative aims to enable AGC to achieve a recycled cullet-to-virgin raw material ratio of up to 56% in automotive glass production [182].

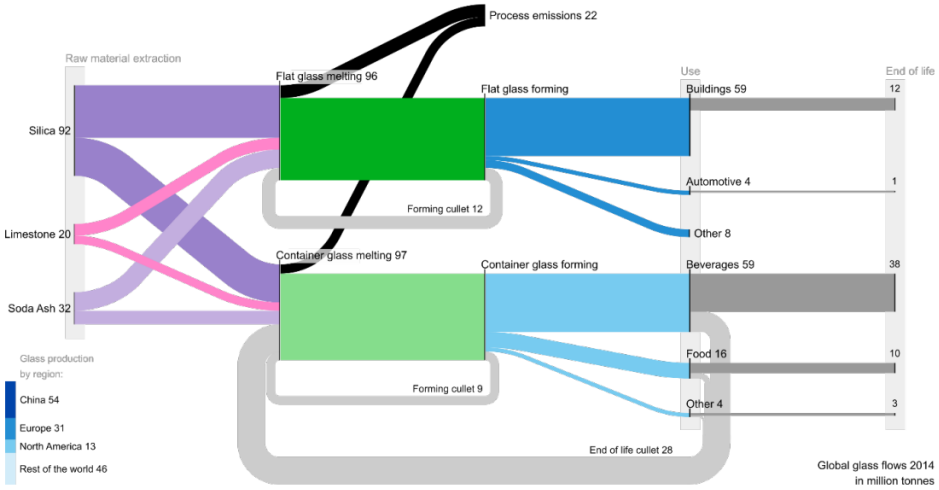


Figure A.2: *Float Glass Material Flow [181]*

Appendix B

Research process

The following diagram contains the full size of the research process diagram used for the development of the project.

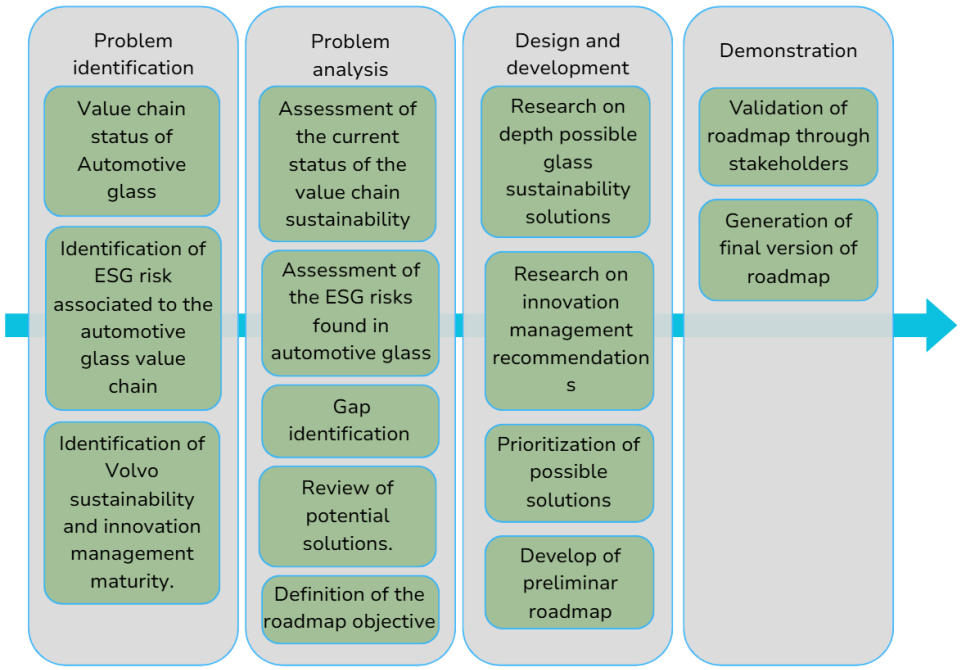


Figure B.1: Research Process for the thesis project

Appendix C

Interview Guide

C.1 Interview Guides

The following guide was used to performed interviews to the different stakeholders in this project,

C.1.1 Sustainable Oriented Interview Guide

1. Can you describe your role in your organization? How would you describe your relation to glass? (Purpose: Identify the interviewee perspective)

2. Under your perspective what makes it harder to make glass more sustainable? (Purpose: Identify rigidities of the industry)

3. Can you give me a recent example of a change in the use or process related to glass that you consider innovative? Which was your role in this innovation? Did it have anything to do with improving the sustainability of the glass? If so, in what way? Is there a difference between innovations that are driven by sustainability and "other" innovations? (Purpose: Understand their current SOI maturity status)

4. When there are new ideas to improve the sustainability of glass, are other departments participating in the discussion? (Purpose: Collaboration dynamics understanding)

5. Do you work with suppliers to develop new glass solutions? or do you adapt according to the market?(Purpose: Maturity level of their

innovation network)

6. (Do you consider that your organization looks actively to find new allies? like Startups, universities, NGO. (Purpose: Maturity level of their innovation)

7. What do you think stops the collaboration between organizations? (Purpose: Understand the blocks to collab)

8. When you are choosing a type of glass or a process, what is more important: cost, technical performance or sustainability impact? How do you measure these factors? (Purpose: Understand what is driving this industry)

9. In your work on glass, what does sustainable glass mean to you? it is more about a less contaminant glass, energy efficient glass or what do you consider it is? (Purpose: Perspective of sustainability, only one layer or more than just environment)

10. How much time or resources do you use for fixing / improving "what you already have"? making the glass more efficient? (Purpose: Evaluate their innovation process and status)

11. Do you think you have space to be more radical or disruptive, thinking outside of the bucket? (Purpose: Evaluate their innovation process and status)

12. If today you have a very disruptive idea for glass, is there a channel and budget to exploit that idea, or your everyday tasks are not allowing it? (Purpose: Evaluate their innovation process and status)

13. Do you feel that sustainability is a real priority now when you take decisions about glass or do you feel there are other important things to comply with before? (Purpose: Understand what is driving this sustainability)

14. How do you keep yourself aware of the new technologies on sustainable glass, do you feel that the company supports this learning or is it because of your personal interest? (Purpose: Understand their capacity to absorb information, how does organization learns).

C.1.2 Specific Stakeholder Additional Interview Guide

C.1.2.1 Manufacturing specialist

1. How is the installation process happening? Is it standardized?
2. Does it generate waste? What happens with the waste? Is it recycled?
3. Are there any specific challenges you are facing in the assembly line when working specifically with glass? If so, how do you deal with them and how often do they occur?

C.1.2.2 Design representative

1. When you work on glass, which is the biggest challenge you see on it, is the supply levels, prices, sustainability, technical requirements, etc?
2. The quality requirements for appearance and performance are based under which information? Do you use the same information from past projects? or do you assess the requirement for each new project?
3. For the glass component, how do you see this component, it is just an exterior layer, or do you see potential on it, that it could do more than it currently does?

C.1.2.3 Recycler representative

1. How is the recycling process for automotive glass conducted? How effective are the current methods, and to what extent are they standardized? Does the process include an initial cleaning stage, and if so, does it involve significant water consumption? (Please consider both laminated and tempered glass.)
2. What volume of automotive glass do you receive for recycling?
3. What proportion of the received glass is reused or sent back to suppliers for reintegration into production processes?
4. Does the recycling process generate waste? If so, how is this waste managed?
5. Who are your main partners in automotive glass recycling?

C.1.2.4 Supplier representative

1. Do you exclusively use internally produced float glass, or do you also source from third-party suppliers? How does the demand for automotive glass compare to the overall production capacity of flat glass?

2. Do you offer solutions involving diboron trioxide (e.g., borosilicate glass)? If so, how are the associated risks managed?

3. Considering only the production stages of automotive glass—from cutting to lamination (tempering) and final inspection—which stage would you identify as having the greatest impact on the sustainability of the product?

4. Among your product portfolio, which product would you consider the most sustainable, and why?

5. Do you recycle post-consumer glass products? If so, how is this process carried out? Do you have the necessary technologies and network to support this?

6. Could you describe the recycling process of laminated and tempered glass collected from end users and processed in your facilities?

7. What recent innovations has your organization developed in automotive glass to reduce emissions and improve energy efficiency throughout the product life cycle?

8. How do you envision the future of the automotive glass industry? In what ways can glass components contribute to the sustainability goals of vehicles, and where is your organization primarily focusing its efforts?

9. What challenges do you encounter when collaborating with OEMs to integrate more sustainable glass solutions?

10. Who are your main raw material suppliers, and to what extent do you rely on local sourcing for key raw materials?

11. Do you primarily use natural soda ash or synthetic soda ash in your production processes?

12. What energy sources are used in the production of float glass?

13. What strategies are being implemented to decarbonize float glass furnaces?

14. Do you require certifications or sustainability standards from your mining suppliers?

15. What is the maximum percentage of recycled glass that can be used without compromising product quality?

16. Is your organization involved in any innovation projects aimed at addressing key sustainability challenges in the glass sector? If so, who are your partners in these initiatives?

17. From your perspective, what actions could Volvo Cars take to support the development of more sustainable automotive glass products?

C.1.2.5 Sustainable third party scheme

Vision, scope, and strategic alignment

1. How would you define the overall aim of the standard? What is your vision for the standard?

2. Which stages of the value chain will be covered by requirements in the standard?

3. Which glass types will fall within scope (e.g., soda-lime only, or also aluminosilicate and borosilicate)? Will fiberglass be included?

4. We understand that the certification scheme will include requirements related to raw material extraction for glass. If so, which materials will be included (e.g., silica sand, soda ash, limestone, diboron trioxide)?

5. Which specific issues have you identified in the extraction of glass raw materials, and are any of these particularly relevant to the automotive industry?

6. Which specific issues have you identified in the glass-making process, and are any of these particularly relevant to automotive glass?

7. Will the standard address end-of-life and circularity actions for automotive glass? Will numeric targets be required (e.g., minimum recycled content percentages)?

8. Which specific issues have you identified in the end-of-life stage of glass, and are any of these particularly relevant to automotive glass?
9. Is the certification scheme developed in collaboration with any other standards (e.g., ISO, IRMA)?

Technical requirements and sustainability challenges

1. Will the standard actively drive improved sustainability (e.g., maximizing efficiency while minimizing ESG impacts), or will it primarily regulate sourcing/outsourcing?
2. How will you evaluate social, political, and economic traceability of the materials? Are there specific methods or structures you plan to apply?
3. How do you plan to ensure that glass is recognized and managed as a strategic resource rather than treated as a simple commodity?

Innovation and regenerative solutions

1. Will certifications include recommendations for potential improvements, or will they be limited to assessing compliance with the standard?
2. Are there planned incentives or requirements for the use of alternative materials (e.g., recycled sand from construction or glass cullet) to reduce primary extraction?
3. Looking ahead, technology continues to evolve. How does the standard plan to respond to technological developments, and how will reviews/updates be conducted?

Governance and institutional relations

1. Will you collaborate with governments in the development and/or implementation of the standard? If so, how? How do you expect the standard to be perceived by national governments?
2. How do you plan to incorporate stakeholder perspectives in the development of the standard (e.g., communities, sustainability NGOs, and others)? Can you share which actors are expected to be most involved?
3. What independent auditing and public transparency mechanisms will the scheme include to prevent conflicts of interest between certified companies and auditing bodies?

Appendix D

Stakeholders Mapping

A brief stakeholder analysis was conducted for the project, resulting in the identification of key internal and external stakeholders, as illustrated in Figure D.1. Internal key stakeholder identified were:

1. **Volvo Materials Engineering Team**
   This team is responsible for the holistic management of materials across all components. Their responsibilities include understanding material extraction processes, as well as their physical and chemical properties. In addition, they define overarching material management strategies, assess potential risks, and monitor emerging trends and market fluctuations.

2. **Volvo Sustainability Team**
   This team is responsible for defining the overall sustainability goals of the organization. They develop the strategies required to achieve these objectives and translate them into actionable targets for each department. Furthermore, they establish processes and guidelines, while also assessing sustainability-related aspects across the organization.

3. **Volvo Glass Design Team**
   The design team is responsible for the development of components and maintains direct interaction with suppliers. They define product requirements, including performance specifications, production constraints, sustainability criteria, and quality standards that must be fulfilled.

4. **Volvo Procurement Team**
   This team is responsible for establishing commercial agreements with suppliers. These agreements encompass compliance with product

requirements, as well as organizational, supply chain, and certification standards. The procurement team ensures that all relevant requirements are incorporated into supplier contracts.

5. **Volvo Legal Team**
   The legal team ensures compliance with all applicable regulatory requirements related to the vehicle. These include regulations concerning product performance, raw material extraction, waste management, and adherence to good production practices.

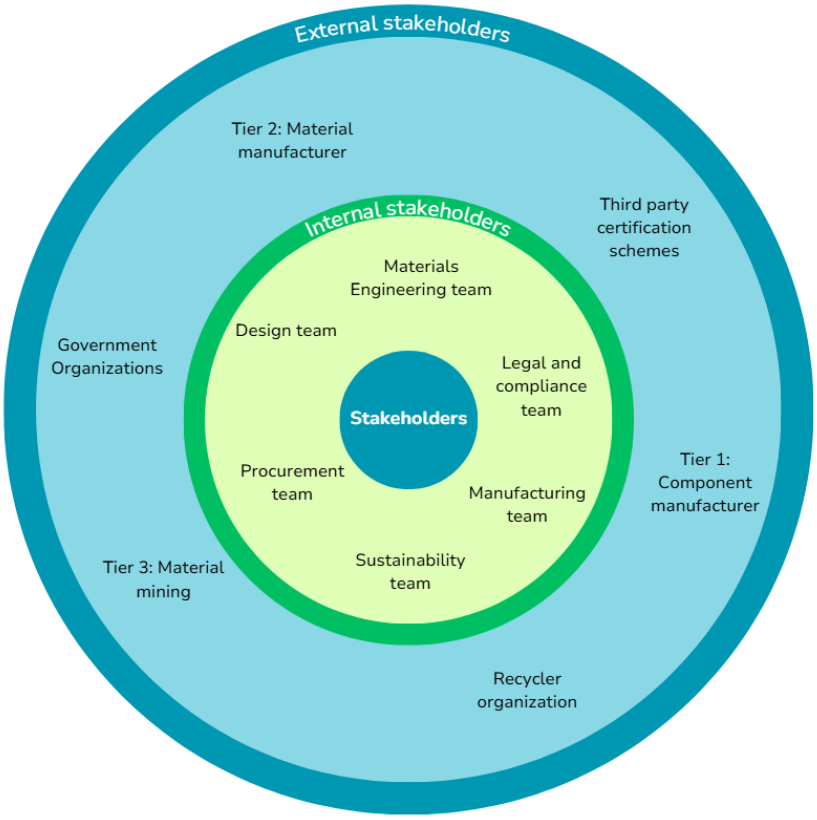


Figure D.1: *Stakeholders map for the project*

Appendix E

Sustainable FMEA (S-FMEA)

List of Risks		Severity	Motivation	Occurance	Motivation	Criticality
Nr	Risk Definition					
1.01	Silicosis disease	5	Silicosis is a fatal occupational disease that exposes suppliers and OEMs to serious legal, human-rights, and reputational risks. Under EU Due Diligence and ESG expectations, OEM responsibility extends upstream.	2	While safety controls exist, enforcement is uneven in silica-producing regions (Asia, Africa, Latin America). Cases still occur, especially in low-cost extraction, this extraction are mostly related to the silica sand mafias which their production is used in locally, specially for construction purposes.	300
1.03	Silica Sand mafias	5	Illegal mining undermines traceability, ESG compliance, and long-term supply security, creating indirect but material risk for OEMs.Through the investigation some suppliers were not keen on being interviewed, this raises suspicion and perhaps imposing higher severity regulation on the supplier regarding silica sand could be a solution.	1	This is geographically specific (e.g. India), not systemic across all sourcing regions. This material is used mostly for local purposes, especially on construction. For the glass flotation there is a quality barrier required to produced good flat glass, reducing the probability of using this type of sand.	300
1.04	Synthetic soda ash production environmental impacts	4	Soda ash production generates significant CO_2 emissions and environmental damage, directly affecting OEM Scope 3 targets and future carbon costs.	3	The majority of soda ash production today is still fossil-fuel-intensive, making this a widespread issue. But is important that this impacts are related to synthetic soda ash, but natural soda ash is more commonly used.	100

Figure E.1: *S-FMEA Part 1*

List of Risks		Severity	Motivation	Occurance	Motivation	Criticality
Nr	Risk Definition					
1.05	**Silica sand mining impact to fauna and flora**	4	Biodiversity damage leads to community opposition, permitting delays, and ESG non-compliance risks.	4	Environmental disruption is inherent to mining activities unless exceptional mitigation measures are applied.	100
1.06	**EU taxonomy on Diboron trioxide**	2	Regulatory classification increases compliance costs and may restrict material use, but alternatives exist.	3	Regulatory pressure is increasing but restrictions are not yet fully prohibitive.	30
2.01	**Energy intense process, use of fossil fuels**	4	Energy use drives cost volatility, high CO_2 emissions, and exposure to carbon taxation; failure to decarbonize threatens competitiveness.	4	Float glass furnaces are structurally dependent on high-temperature fossil energy today. While the glass sector is working on decarbonization plans with targets of carbon neutrality from 2040 to 2050. The glass industry sector is dependent on natural gas and European market have been affected by regional geopolitical dispute.	100
2.02	**Oligopoly market**	4	Supplier concentration increases price leverage, reduces supply resilience, and limits OEM strategic freedom.	4	Market consolidation is already happening and entry barriers are extremely high.	100
2.03	**Risk of PFAS use in hydrophobic glass (Coating application).**	4	PFAS regulation in the future can force design changes, product bans, or liability exposure, directly impacting product approval.	2	PFAS are not used universally but are common in advanced coatings (hydrophobic, PV glass).	75
2.04	**IP Rights**	2	OEMs are locked into supplier IP, limiting recyclability innovation, supplier switching, and circular economy progress.	4	Glass chemistry, coatings, and furnace technology are tightly protected across the industry. But this situation is systematically like that in the industry.	30
2.05	**Trade tension**	4	Trade barriers, tariffs, or political restrictions affect supply continuity and cost.	2	Geopolitical tensions are recurring but not consistently disruptive.	75

Figure E.2: *S-FMEA Part 2*

List of Risks		Severity	Motivation	Occurance	Motivation	Criticality
Nr	Risk Definition					
5.01	Waste from automotive aftermarket	2	Drives raw material demand, waste generation, and insurance/environmental costs. Issue is not fully in Volvo's scope.	5	Glass breakage and replacement are inherent to vehicle operation.	30
5.02	Low levels of circularity	4	Contradicts OEM sustainability commitments and exposes companies to future regulatory and reputational risk.	4	Recycling infrastructure for laminated automotive glass is still limited, KPIs on glass industy suppliers show ratios between 1 to 8%	100
5.03	Automotive Glass quality	2	Safety regulations slow innovation, preventing rapid adoption of circular solutions. But this constraint is mostly applied to windshields and the industry is used to it.	4	Homologation applies to every vehicle platform, making this a systemic constraint.	30
5.04	EU Provisional agreements	4	Future Non-compliance can result in fines, market access loss, and OEM reputational damage.	2	Current dismantling, sorting, and recycling capacity is between 60 to 70% based on the latest Volvo's model. On the other hand, the glass is recycled but degraded, so is not currently a risk.	75

Figure E.3: *S-FMEA Part 3*

Appendix F

Materiality Matrix

A zoomed version of the materiality matrix 4 quadrants derived are presented. Top-left quadrant in Figure F.1, top-right quadrant in Figure F.2, bottom-left quadrant in Figure F.3 and bottom-right quadrant in Figure F.4.



Figure F.1: *Top-Left Quadrant: Low impact, high effort*

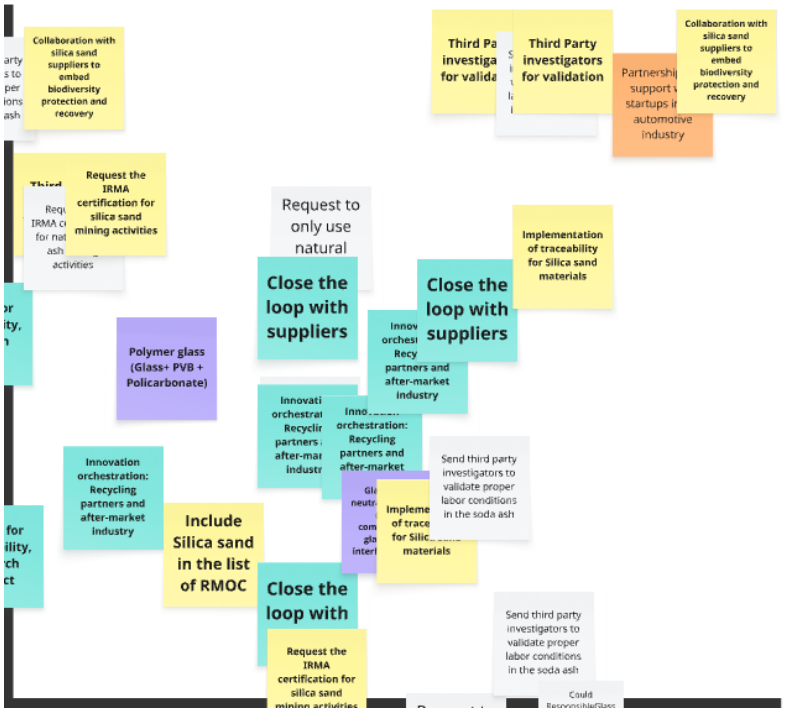


Figure F.2: *Top-Right Quadrant: High impact, high effort*

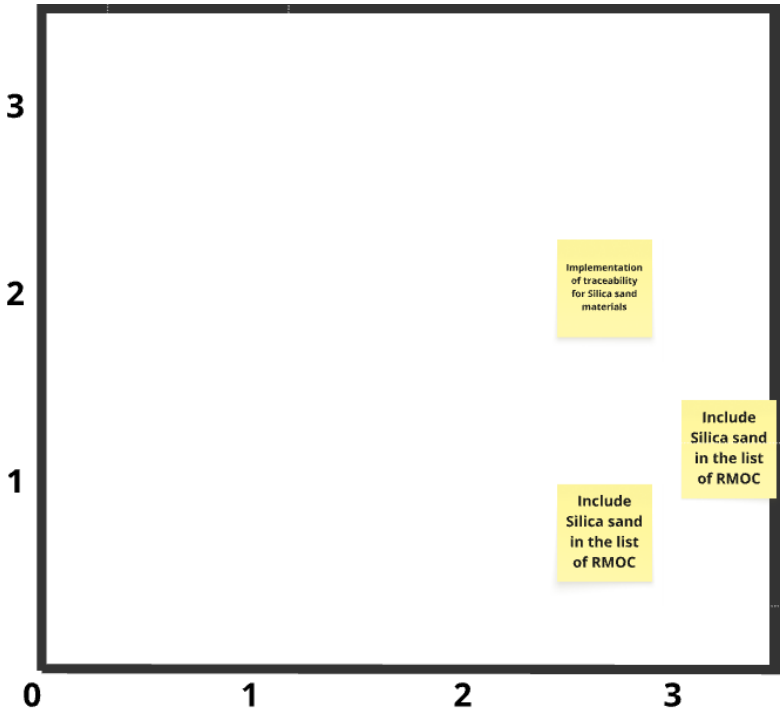


Figure F.3: *Bottom-Left Quadrant: Low impact, Low effort*

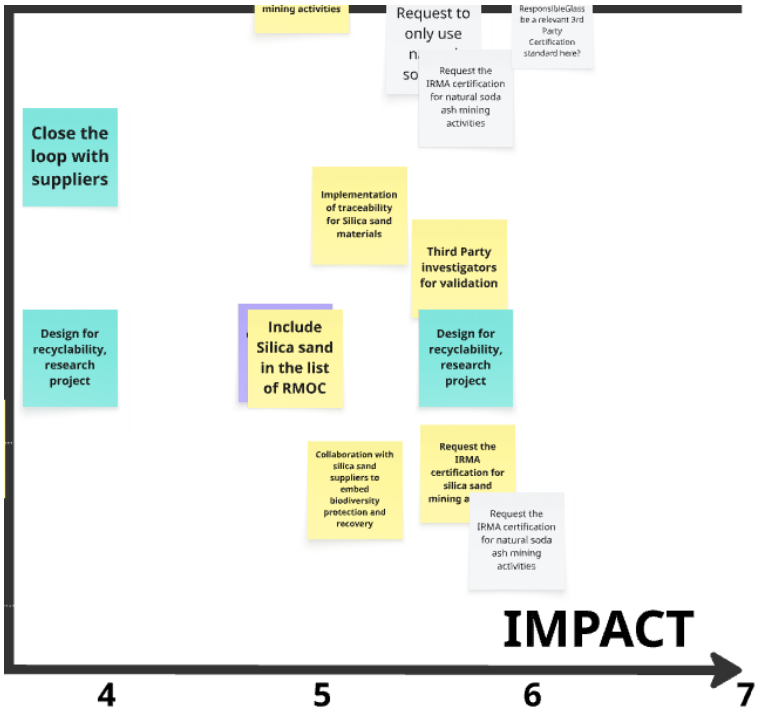


Figure F.4: *Bottom-Right Quadrant: High impact, Low effort*

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